CHAPTER 5 Molecular Biology and Genetics

SYLLABUS

Molecular Structure of Genes and Chromosomes; Mutations and Mutagenesis; Nucleic Acid Replication, Transcription, Translation and their Regulatory Mechanisms in Prokaryotes and Eukaryotes; Mendelian Inheritance; Gene Interaction; Complementation; Linkage, Recombination and Chromosome Mapping; Population Genetics; Extra Chromosomal Inheritance; Microbial Genetics (Plasmids, Transformation, Transduction, Conjugation); Horizontal Gene Transfer and Transposable Elements; RNA Interference; DNA Damage and Repair; Chromosomal Variation; Molecular Basis of Genetic Diseases.

CHAPTER ANALYSIS

Topic	GATE 2010	GATE 2011	GATE 2012	GATE 2013	GATE 2014	GATE 2015	GATE 2016	GATE 2017	GATE 2018	GATE 2019
Molecular Structure of Genes and Chromosomes	1		1	1	4			2	3	
Mutations and Mutagenesis	1	1	1							1
Nucleic Acid Replication, Transcription, Translation and their Regulatory Mechanisms in Prokaryotes and Eukaryotes	4	4	1	4	1	3	1	3	2	1
Mendelian Inheritance	1			1	1	1	1			2
Gene Interaction					1					
Complementation			1							
Linkage, Recombination and Chromosome Mapping										
Population Genetics				1	2	1				
Extra Chromosomal Inheritance	1		1							
Microbial Genetics: Plasmids, Transformation, Transduction, Conjugation	1	2	2			1		1		1
Horizontal Gene Transfer and Transposable Elements				1						
RNA Interference							1		1	
DNA Damage and Repair			1							
Chromosomal Variation		1	1	1						
Molecular Basis of Genetic Diseases		2	1	1			2	1	1	

QUESTIONS

1. The G + C content of bacteriophage M13 double stranded DNA is 68%. What would you expect the G + C content of its mRNA?

(A) About 68%
(B) About 34%
(C) About 32%
(D) About 86%

(GATE 2000; 1 Mark)

2. DNA helicases catalyze the reaction
   (A) DNA supercoiling.
   (B) DNA unwinding.
   (C) cleavage of DNA helix.
   (D) interconversion of DNA topoisomers.

(GATE 2000; 1 Mark)

3. Rho-dependent and rho-independent transcription termination mechanisms operate in prokaryotes. Rho independent termination mechanism involves
   (A) binding of the rho protein upstream of the termination element.
   (B) no protein factors and only RNA secondary structure and run of 'U's.
   (C) presence of UGA or UAA stop codons.
   (D) binding of accessory factors at termination signal.

(GATE 2000; 1 Mark)

4. In eukaryotes, the ribosomal RNA genes are transcribed by
   (A) reverse transcriptase.
   (B) RNA dependent RNA polymerase.
   (C) RNA polymerase I.
   (D) RNA polymerase III.

(GATE 2000; 1 Mark)

5. The term 'protein splicing' refers to
   (A) removal of intervening sequences between the genes.
   (B) splicing out of introns from RNA.
   (C) removal of intervening protein sequences from the translated protein.
   (D) joining (splicing) of two different gene products to generate a novel protein.

(GATE 2000; 1 Mark)

6. PKU is one of the best-known hereditary disorders in amino acid metabolism. The defect is attributed to a lesion in one of the following enzymatic activities:
   (A) Phenylalanine ammonia lyase
   (B) Phenylalanine hydroxylase
   (C) Tyrosine hydroxylase
   (D) Phenylalanine transaminase

(GATE 2000; 1 Mark)

7. The double stranded DNA molecule of a virus was found by electron microscopy to have a length of 34 μm.
   (i) How many nucleotide pairs are there in one molecule?
   (ii) How many complete turns of the two polynucleotide chains are present in such a double helix?

(GATE 2000; 2 Marks)

8. Efficient expression of a heterologous protein product is influenced by
   (A) transcriptional efficiency.
   (B) copy number of the plasmid.
   (C) codon bias.
   (D) all of the above.

(GATE 2001; 1 Mark)

9. The fundamental feature of the genetic code which allows the expression of a protein in any host is its
   (A) triplet nature. (B) universality.
   (C) degeneracy. (D) redundancy.

(GATE 2001; 1 Mark)

10. The major groove of DNA is lined by
   (A) N3 of purine and N1 of pyrimidine.
   (B) N7 of purine and O2 of pyrimidine.
   (C) N7 of purine and C4 of pyrimidine.
   (D) None of the above.

(GATE 2001; 1 Mark)

11. Reverse genetics means
   (A) finding the function of an ORF.
   (B) finding the gene responsible for a trait.
   (C) RNA dependent DNA synthesis.
   (D) convening somatic cell to a germ cell.

(GATE 2001; 1 Mark)

12. Mention an important post translational modification absent in prokaryotes making them unsuitable hosts for expressing human genes.

(GATE 2001; 1 Mark)

13. RNA is very much susceptible to hydrolysis in alkali because
   (A) it contains uracil residues in its structure
   (B) its 2′–OH group participate in intramolecular cleavage of phosphodiester backbone
   (C) cleavage occurs in the glycosylic bonds of purine bases
   (D) cleavage occurs in the glycosylic bonds of pyrimidine bases

(GATE 2002; 1 Mark)

14. The full-length coding sequence of an eukaryotic gene was expressed in bacteria and the protein was purified. However, in the functional assay, no activity was detected for the purified protein. The reason could be
   (A) the host bacteria produced an enzyme that inhibited the activity of the expressed eukaryotic protein.
   (B) the purified protein was contaminated with bacteria.

(C) the host bacteria did not produce the essential co-factors.

(D) no post-translational modification on the protein expressed in bacteria.

(GATE 2003; 2 Marks)

15. Some of the genes from viruses introduced into plants in fully functional form often exhibit Mendelian inheritance because

(A) the genes are stably integrated in chromosomes.

(B) the genes are stably maintained in vectors.

(C) the genes are co-expressed with chromosomal genes.

(D) the genes are not interrupted by introns.

(GATE 2004; 2 Marks)

16. siRNA(s) interfere at

(A) transcriptional level.

(B) post-transcriptional level.

(C) DNA replication level.

(D) translational level.

(GATE 2007; 1 Mark)

17. Mobile genetic elements present in human genome are

P. Long interspersed elements (LINEs)

Q. Short interspersed elements (SINEs)

R. P elements

S. IS elements

(A) Q, R (B) P, Q

(C) P, R (D) Q, S

(GATE 2007; 2 Marks)

18. Match the terms in Group I with their possible explanation in Group II.

Group I	Group II
P. Orthologs	1. A cell or an organism, having foreign gene.
Q. Paralogs	2. The complement of a protein expressed by a genome.
R. Proteome	3. Genes from different species related to each other.
S. Transgenic	4. Genes from same species related to each other.

(A) P–2; Q–4; R–1; S–3 (B) P–4; Q–3; R–2; S–1

(C) P–3; Q–4; R–2; S–1 (D) P–1; Q–2; R–3; S–4

(GATE 2008; 2 Marks)

19. Which of the following statements are true with respect to a special complex called 'dicer'?

P. It consists of deoxyribonuclease and DNA fragments.

Q. It consists of ribonuclease and RNA fragments.

R. It is involved in gene silencing.

S. It triggers apoptosis.

(A) P, R (B) Q, R

(C) P, S (D) Q, S

(GATE 2008; 2 Marks)

20. Caspases are involved in the process of

(A) DNA replication.

(B) mutation and recombination.

(C) antibody synthesis.

(D) apoptosis.

(GATE 2009; 1 Mark)

21. Ames test is used to determine

(A) the mutagenicity of a chemical.

(B) carcinogenicity of a chemical.

(C) both mutagenicity and carcinogenicity of a chemical.

(D) toxicity of a chemical.

(GATE 2010; 1 Mark)

22. Peptidyl transferase activity resides in

(A) 16S rRNA. (B) 23S rRNA.

(C) 5S rRNA. (D) 28S rRNA.

(GATE 2010; 1 Mark)

23. In transgenics, alterations in the sequence of nucleotide in genes are due to

P. Substitution Q. Deletion

R. Insertion S. Rearrangement

(A) P and Q (B) P, Q and R

(C) Q and R (D) R and S

(GATE 2010; 1 Mark)

24. During transcription,

(A) DNA gyrase introduces negative supercoils and DNA topoisomerase I removes negative supercoils.

(B) DNA topoisomerase I introduces negative supercoils and DNA gyrase removes negative supercoils.

(C) both DNA gyrase and DNA topoisomearse I introduce negative supercoils.

(D) both DNA gyrase and DNA topoisomearse I remove negative supercoils.

(GATE 2010; 1 Mark)

25. An example for template independent DNA polymerase is

(A) DNA Polymerase I.

(B) RNA polymerase.

(C) Terminal deoxynucleotidyl transferase.

(D) DNA polymerase III.

(GATE 2010; 1 Mark)

26. Nuclease-hypersensitive sites in the chromosomes are sites that appear to be

(A) H2 and H4 histone free.
(B) H1 and H2 histone free.
(C) H3 and H4 histone free.
(D) nucleosome free.

(GATE 2010; 1 Mark)

27. The formation of peptide cross links between adjacent glycan chains in cell wall synthesis is called

(A) transglycosylation.
(B) autoglycosylation.
(C) autopeptidation.
(D) transpeptidation.

(GATE 2010; 1 Mark)

28. Determine the correctness or otherwise of the following Assertion (*a*) and the Reason (*r*).

Assertion: Cytoplasmic male sterility (cms) is invariably due to defect(s) in mitochondrial function.

Reason: cms can be overcome by pollinating a fertility restoring (*Rf*) plant with pollen from a non cms plant.

(A) Both (*a*) and (*r*) are true and (*r*) is the correct reason for (*a*).
(B) Both (*a*) and (*r*) are true and (*r*) is not the correct reason for (*a*).
(C) (*a*) is false but (*r*) is true.
(D) (*a*) is true but (*r*) is false

(GATE 2010; 2 Marks)

29. True breeding *Drosophila* flies with curved wings and dark bodies were mated with true breeding short wings and tan body *Drosophila*, The F_1 progeny was observed to be with curved wings and tan body. The F_1 progeny was again allowed to breed and produced flies of the following phenotype, 45 curved wings tan body, 15 short wings tan body, 16 curved wings dark body and, 6 short wings dark body. The mode of inheritance is

(A) typical Mendelian with curved wings and tan body being dominant.
(B) typical non-Mendelian with curved wings and tan body not following any pattern.
(C) Mendelian with suppression of phenotypes.
(D) Mendelian with single crossover.

(GATE 2010; 2 Marks)

30. Diploid *Drosophila* has eight chromosomes. Which one of the following terms should not be used to describe *Drosophila* with sixteen numbers of chromosomes?

(A) Polyploid **(B)** Aneuploid
(C) Euploid **(D)** Tetraploid

(GATE 2011; 1 Mark)

31. Molecular chaperones are class of proteins that facilitate

(A) the proper folding of newly synthesized proteins.
(B) unfolding of newly synthesized proteins.
(C) degradation of newly synthesized proteins.
(D) targeting of newly synthesized proteins.

(GATE 2011; 1 Mark)

32. The protein in eukaryotes which is subjected to degradation undergoes

(A) phosphorylation.
(B) carboxylation.
(C) ubiquitation.
(D) methylation.

(GATE 2011; 1 Mark)

33. Determine the correctness or otherwise of the following Assertion (*A*) and the Reason (*R*).

Assertion: N-methyl-N′-nitro-N-nitrosoguanidine (NTG) is an effective chemical mutagen.

Reason: Mutations induced by NTG mainly are the GC → AT transitions.

(A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(B) Both (*A*) and (*R*) are true but (*R*) is not the correct reason for (*A*).
(C) (*A*) is true but (*R*) is false.
(D) (*A*) is false but (*R*) is true.

(GATE 2011; 2 Marks)

34. Determine the correctness of the following statements.

I. Enhancer sequences are those DNA sequences that are involved in increasing the rate of DNA replication.

II. Enhancer sequences work by binding with eukaryotic gene activator factors.

(A) Only I is true.
(B) Only II is true.
(C) Both I and II are true.
(D) Both I and II are false.

(GATE 2011; 2 Marks)

35. Match the terms in Group I with their associated functions in Group II.

Group I	Group II
P. Shine-Dalgarno sequences	1. Aminoacylation of tRNA
Q. Leucine zipper	2. Gene silencing
R. Aminoacyl tRNA synthetase	3. Ribosome binding and facilitation of translation initiation
S. RNA interference (RNAi)	4. Transcription factors

(A) P–3; Q–4; R–1; S–2
(B) P–4; Q–3; R–2; S–1
(C) P–2; Q–3; R–1; S–4
(D) P–3; Q–2; R–4; S–1

(GATE 2011; 2 Marks)

36. Determine the correctness or otherwise of the following Assertion (*A*) and the Reason (*R*).

Assertion: Isopropylthiogalactoside (IPTG) is a gratuitous inducer of lactose operon.

Reason: Gratuitous inducers are chemical analogs which behave like natural inducer but they do not serve as substrate for the enzymes that are subsequently synthesized.

(A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(B) Both (*A*) and (*R*) are true but (*R*) is not the correct reason for (*A*).
(C) (*A*) is true but (*R*) is false.
(D) (*A*) is false but (*R*) is true.

(GATE 2011; 2 Marks)

37. Which of the following characteristics with respect to bacterial DNA polymerase III are true?

P. Initiation of chain synthesis
Q. 5′→ 3′ polymerization
R. 3′→ 5′ exonuclease activity
S. 5′→ 3′ exonuclease activity

(A) P and Q only (B) Q and R only
(C) Q and S only (D) P and S only

(GATE 2011; 2 Marks)

Common Data for Questions 38 and 39: Red-green color blindness is inherited as a recessive X-linked trait.

38. What will be the probability of having the color-blind son to a woman with phenotypically normal parents and a color-blind brother, and married to a normal man? (Assume that she has no previous children)

(A) 100 % (B) 50 %
(C) 25% (D) 12.5 %

(GATE 2011; 2 Marks)

39. What will be the probability of having the color-blind daughter to a phenotypically normal woman, who already had one color-blind son, and is married to a color-blind man?

(A) 75 % (B) 50 %
(C) 25 % (D) 15%

(GATE 2011; 2 Marks)

40. In mismatch correction repair, the parental DNA strand is distinguished from the daughter strand by

(A) acetylation. (B) phosphorylation.
(C) methylation. (D) glycosylation.

(GATE 2012; 1 Mark)

41. A protein is phosphorylated at a serine residue. A phosphomimic mutant of the protein can be generated by substituting that serine with

(A) glycine. (B) alanine.
(C) aspartate. (D) threonine.

(GATE 2012; 1 Mark)

42. A truncated polypeptide is synthesized due to a nonsense mutation. Where would you introduce another mutation to obtain a full-length polypeptide?

(A) Ribosomal protein gene
(B) Transfer RNA gene
(C) DNA repair gene
(D) Ribosomal RNA gene

(GATE 2012; 1 Mark)

43. The direction of shell coiling in the snail *Limnaea peregra* is a classic example of

(A) chromosomal inheritance.
(B) extra-chromosomal inheritance.
(C) chromosomal translocation.
(D) homologous recombination.

(GATE 2012; 1 Mark)

44. Synteny refers to

(A) gene duplication from a common ancestor.
(B) a tree representation of related sequences.
(C) the extent of similarity between two sequences.
(D) local conservation of gene order.

(GATE 2012; 1 Mark)

45. Which one of the following DNA sequences carries an invert repeat?

(A) ATGAGCCCCGAGTA
TACTCGGGGCTCAT
(B) ATGAGCCGGCTCTA
TACTCGGCCGAGAT
(C) ATGAGCCGAGCCTA
ACTCGGCTCGGAT
(D) ATGAGCCTATGGTA
TACTCGGATACCAT

(GATE 2012; 2 Marks)

46. Match the terms in Group I with the ploidy in Group II.

Group I	Group II
P. Disome	1. $2n + 1$
Q. Monosome	2. $2n - 1$
R. Nullisome	3. $n - 1$
S. Trisome	4. $n + 1$

(A) P–4; Q–2; R–3; S–1 (B) P–4; Q–3; R–1; S–2
(C) P–2; Q–3; R–4; S–1 (D) P–1; Q–4; R–3; S–2

(GATE 2012; 2 Marks)

47. A disease is inherited by a child with a probability of 1/4. In a family with two children, the probability that exactly one sibling is affected by this disease is

(A) 1/4 (B) 3/8
(C) 7/16 (D) 9/16

(GATE 2012; 2 Marks)

Statement for Linked Answer Questions 48 and 49: An *E. coli* cell of volume 10^{-12} cm^3 contains 60 molecules of lac-repressor. The repressor has a binding affinity (K_d) of 10^{-8} M and 10^{-9} M with and without lactose respectively, in the medium.

48. The molar concentration of the repressor in the cell is

(A) 0.1 nM (B) 1 nM
(C) 10 nM (D) 100 nM

(GATE 2012; 2 Marks)

49. Therefore, the lac operon is

(A) repressed and can only be induced with lactose.
(B) repressed and cannot be induced with lactose.
(C) not repressed.
(D) expressed only when glucose and lactose are present.

(GATE 2012; 2 Marks)

50. Under alkaline conditions, DNA is more stable than RNA because

(A) RNA forms secondary structures.
(B) RNA is a single stranded molecule.
(C) RNA has uracil in place of thymidine.
(D) RNA is susceptible to hydrolysis.

(GATE 2013; 1 Mark)

51. Which one of the following modifications is common to both protein and DNA?

(A) SUMOylation. (B) nitrosylation.
(C) methylation. (D) ubiquitination.

(GATE 2013; 1 Mark)

52. Which one of the following amino acids in proteins does not undergo phosphorylation?

(A) Ser (B) Thr
(C) Pro (D) Tyr

(GATE 2013; 1 Mark)

53. In nature, the horizontal gene transfer across bacteria is mediated by

(A) gene cloning followed by transformation.
(B) conjugation and transformation.
(C) conjugation only.
(D) transformation only.

(GATE 2013; 1 Mark)

54. Of the two diploid species, species I has 36 chromosomes and species II has 28 chromosomes. How many chromosomes would be found in an allotriploid individual?

(A) 42 or 54 (B) 46 or 50
(C) 74 or 86 (D) 84 or 108

(GATE 2013; 1 Mark)

55. The RNA primer synthesized during the replication process in bacteria is removed by

(A) DNA gyrase.
(B) primase.
(C) DNA polymerase I.
(D) DNA polymerase II.

(GATE 2013; 1 Mark)

56. Hypophosphatemia is manifested by an X-linked dominant allele. What proportion of the offspring from a normal male and an affected heterozygous female will manifest the disease?

(A) 1/2 sons and 1/2 daughters.
(B) all daughters and no sons.
(C) all sons and no daughters.
(D) 1/4 daughters and 1/4 sons.

(GATE 2013; 1 Mark)

57. Match the entries in Group I with the entries in Group II.

Group I	Group II
P. RNAse P	1. Polyadenylation
Q. RNase H	2. Splicing
R. snRNAs	3. Ribozymes
S. CstF	4. DNA-RNA hybrids

(A) P–3; Q–4; R–2; S–1
(B) P–4; Q–3; R–2; S–1
(C) P–3; Q–2; R–1; S–4
(D) P–2; Q–4; R–1; S–3

(GATE 2013; 2 Marks)

58. In a genetic study, 80 people were found to have alleles for polydactyly. Only 36 of them were polydactylous. What is the extent of penetrance percentage?

(GATE 2013; 2 Marks)

59. The 4-amino or 4-keto group of pyrimidine bases is located in the

(A) major groove of the double stranded DNA.
(B) minor groove of the double stranded DNA.
(C) minor groove of the B form DNA but not the A form DNA.
(D) major groove of the B form DNA but not the A form DNA.

(GATE 2014; 1 Mark)

60. The 5′ ends of the mature forms of the prokaryotic mRNAs and tRNAs are

(A) a triphosphate group in mRNAs and a monophosphate group in tRNAs.

(B) triphosphate groups in both mRNAs and tRNAs.

(C) monophosphate groups in both mRNAs and tRNAs.

(D) a monophosphate group in mRNAs and a triphosphate group in tRNAs.

(GATE 2014; 1 Mark)

61. In a relatively large but finite and closed population of sexually reproducing diploid organisms, the frequency of homozygous genotype *PP* changes from 0.40 to 0.50 and that of *pp* changes from 0.40 to 0.41 in a span of 10 generations. Which of the following is the most likely cause for the above change in frequency of the *PP* genotype?

(A) Non-random mating

(B) Random genetic drift

(C) Selection

(D) Combination of non-random mating and random genetic drift

(GATE 2014; 2 Marks)

62. Topological winding number of a 2.0 kb covalently closed circular DNA was found to be 191 with a writhing number of –4. Hence, its linking number and the number of base pair per turn when the molecule is laid flat on the surface is _____and _____, respectively.

(A) 187, 10.69 **(B)** 195, 10.25

(C) 200, 10.00 **(D)** 187, 10.50

(GATE 2014; 2 Marks)

63. Consider a population of 10,000 individuals, of which 2500 are homozygotes (*PP*) and 3000 are heterozygotes (*Pp*) genotype. The frequency of allele *p* in the population is _____.

(GATE 2014; 2 Marks)

64. For their efficient translation, eubacterial mRNAs possess a Shine-Dalgarno sequence for its recognition by an anti-Shine-Dalgarno sequence (ASD) in the ribosomes. The correct statement is

(A) ASD is present in 5S rRNA.

(B) ASD is present in 23S rRNA.

(C) ASD is present in 16S rRNA.

(D) ASD is formed by the interaction of the 16S rRNA with the 23S rRNA upon docking of the 50S subunit on the 30S subunit of the ribosomes.

(GATE 2014; 2 Marks)

65. In a genetic cross between the genotypes *WWXX* and *wwxx*, the following phenotypic distributions were observed among the F_2 progeny: *WX*, 562; *wx*, 158; *Wx*, 38; and *wX*, 42. Likewise, a cross between *XXYY* and *xxyy* yielded the following results: *XY*, 675; *xy*, 175; *Xy*, 72; and *xY*, 78. Similarly, a cross between *WWYY* and *wwyy* yielded: *WY*, 292; *wy*, 88; *Wy*, 12; and *wY*, 8. In all the genotypes, capital letters denote the dominant allele. Assume that the F_1 progeny were self-fertilized in all three crosses. Also, double cross-over does not occur in this species. Which of the following is correct?

(A) Relative position: *W–X–Y* Distances: *W–X* = 5 map units, X–Y = 17 map units

(B) Relative position: *X–Y–W* Distances: *X–Y* = 15 map units, *Y–W* = 11 map units

(C) Relative position: *Y–W–X* Distances: *Y–W* = 5 map units, *W–X* = 11 map units

(D) Relative position: *X–W–Y* Distances: *X–W* = 5 map units, *W–Y* = 10 map units

(GATE 2014; 2 Marks)

66. Which of the following statements with respect to the orientation of the nitrogenous bases to the pentose sugars, and the puckering of the sugar, is correct?

(A) Anti, and 2′-endo in A form DNA

(B) Anti, and 2′-endo in B form DNA

(C) Syn, and 3′-endo in A form DNA

(D) Syn, and 3′ -endo in B form DNA

(GATE 2014; 2 Marks)

67. A dioecious plant has XX sexual genotype for female and XY for male. After double fertilization, what would be the genotype of the embryos and endosperms?

(A) 100% ovules will have XXX endosperm and XX embryo.

(B) 100% ovules will have XXY endosperm and XY embryo.

(C) 50% ovules will have XYY endosperm and XY embryo, while other 50% will have XXY endosperm and YY embryo.

(D) 50% ovules will have XXX endosperm and XX embryo, while the other 50% will have XXY endosperm and XY embryo.

(GATE 2014; 2 Marks)

68. How many 3-tuples are possible for the following amino acid sequence?

MADCMWDISEASE

(A) 4 **(B)** 5

(C) 11 **(D)** 12

(GATE 2015; 1 Mark)

69. How many different protein sequences of 100 residues can be generated using 20 standard amino acids?

(A) 100^{20} **(B)** 100×20

(C) 20^{100} **(D)** $100! \times 20!$

(GATE 2015; 1 Mark)

70. An isolated population on an island has the following genotypic frequencies:

Genotype	*AA*	*Aa*	*aa*
Frequency	0.3	0.4	0.3

Assuming that there are only two alleles (*A* and *a*) for the gene, the genotypic frequency of *AA* in the next generation will be _____.

(GATE 2015; 2 Marks)

71. A heterozygous tall plant (*Tt*) was crossed with a homozygous dwarf plant (*tt*). The resultant seeds were collected. If five seeds are chosen at random, then the probability (in %) that exactly two of these seeds will yield dwarf plants is _____.

(GATE 2015; 2 Marks)

72. An *in vitro* translation system can synthesize peptides in all three reading frames of the RNA template. When 5′ – UCUCUCUC—$(UC)_n$—UCUCUCUC – 3′ was used as the template in this *in vitro* translation system, the synthesized peptides contained 50% each of serine and leucine. When 5′ – CCUCCUCCU—$(CCU)_n$—CCUCCU – 3′ was used as the template, the synthesized peptides contained 33.3% each of serine, leucine, and proline. Deduce the codon for proline.

(A) UCU **(B)** CUC
(C) CCU **(D)** UCC

(GATE 2015; 2 Marks)

73. Cytoplasmic extract from the wild type strain of a bacterium has the ability to convert a colorless substrate (S) to a colored product (P) via three colorless intermediates X, Y and Z, in that order. Each step of the pathway involves a specific enzyme coded by a distinct gene. Four mutant strains (a^-, b^-, c^-, d^-) were isolated, whose extracts are incapable of producing the colored product in the presence of S. In a series of experiments, extracts from the individual mutants were incubated with X, Y, or Z and scored for color development. The data are summarized in the table below. (Yes: color developed, No: no color developed)

	Compounds		
Mutants	X	Y	Z
a^-	No	No	No
b^-	No	Yes	Yes
c^-	Yes	Yes	Yes
d^-	No	No	Yes

Based on the data, which one of the following is the correct order of enzymes involved in the pathway?

(A) $S \xrightarrow{d} X \xrightarrow{c} Y \xrightarrow{b} Z \xrightarrow{a} P$
(B) $S \xrightarrow{a} X \xrightarrow{d} Y \xrightarrow{b} Z \xrightarrow{c} P$
(C) $S \xrightarrow{b} X \xrightarrow{a} Y \xrightarrow{c} Z \xrightarrow{d} P$
(D) $S \xrightarrow{c} X \xrightarrow{b} Y \xrightarrow{d} Z \xrightarrow{a} P$

(GATE 2015; 2 Marks)

74. Which one of the following is not a therapeutic agent based on nucleic acid for the treatment of genetic disorders?

(A) Antisense oligonucleotide
(B) Ribozyme
(C) Aptamer
(D) Avidin

(GATE 2016; 1 Mark)

75. Based on their function, find the odd one out.

(A) miRNA **(B)** siRNA
(C) shRNA **(D)** snRNA

(GATE 2016; 1 Mark)

76. Which set of the following events occurs during the elongation step of translation?

P. Attachment of mRNA with the smaller subunit of ribosome
Q. Loading of correct aminoacyl-tRNA into the A site.
R. Formation of a peptide bond between the amino acyl-tRNA in the A site and the peptide chain that is attached to the peptidyl-tRNA in the P site.
S. Dissociation of the ribosomal subunits.
T. Translocation of peptidyl-tRNA from the A site to the P site of the ribosome.

(A) P, Q and R **(B)** P, Q and T
(C) Q, R and T **(D)** R, S and T

(GATE 2016; 2 Marks)

77. A DNA sequence, 5′–ATGGACGTGCTTCCCAAAG-CATCGGGC–3′, is mutated to obtain

P. 5′– ATGGACGTGCTTCaCAAAGCATCGGGC –3′
Q. 5′– ATGGACGTGCTTCCCgAAAGCATCGGGC –3′
R. 5′– ATGGACGTGCTTCC–AAAGCATCGGGC –3′
S. 5′– ATGGACGTGCTTCCCAAtGCATCGGGC –3′
T. 5′– ATGGACGaGCTTCCCAAAGCATCGGGC –3′

[Point mutations are shown in the lower case or '–' within the sequences]

Which of the above mutant sequences do not have frameshift?

(A) P, Q and S **(B)** P, S and T
(C) Q, R and S **(D)** Q, S and T

(GATE 2016; 2 Marks)

78. Match the type of chromosomal inheritance (Column I) with the corresponding genetic disease or trait (Column II).

Column I	Column II
P. Autosomal recessive inheritance	1. Huntington disease
Q. Autosomal dominant inheritance	2. Hairy ears
R. X-linked inheritance	3. Cystic fibrosis
S. Y-linked inheritance	4. Hemophilia

(A) P–1; Q–4; R–3; S–2
(B) P–4; Q–3; R–2; S–1
(C) P–3; Q–1; R–4; S–2
(D) P–4; Q–2; R–3; S–1

(GATE 2016; 2 Marks)

79. A crossing was performed between the genotypes *DdEeFfgg* and *ddEeFfGg*. Assuming that the allelic pairs of all genes assort independently, the proportion of progeny having the genotype *ddeeffgg* is expected to be _____%.

(GATE 2016; 2 Marks)

80. If the nucleotide composition (%) of a viral genome is A = 10, U = 20, C = 40, and G = 30, which one of the following is this genome?

(A) Double stranded RNA
(B) Single stranded RNA
(C) Single stranded DNA
(D) Double stranded DNA

(GATE 2017; 1 Mark)

81. During protein synthesis, tRNAs are not involved in

(A) charging.
(B) initiation.
(C) elongation.
(D) termination.

(GATE 2017; 1 Mark)

82. The transcription factor X binds a 10 base pair DNA stretch, In the DNA of an organism, X was found to bind at 20 distinct sites. An analysis of these 20 binding sites showed the following distribution:

Base	Position in the binding site									
	1	2	3	4	5	6	7	8	9	10
A	11	0	0	0	16	2	4	0	4	3
T	3	0	19	0	1	3	4	20	2	4
G	4	20	0	0	2	4	6	0	12	2
C	2	0	1	20	1	11	6	0	2	11

What is the consensus sequence for the binding site of X?

(A) NGTCNNNTNN
(B) AGTCACNTGC
(C) CACCTANCTG
(D) ANNNACGNGC

(GATE 2017; 1 Mark)

83. A bacterium has a genome of size 6 million base pairs: If the average rate of DNA synthesis is 1000 base pairs/second, the time taken (in minutes) for replication of the genome will be _____.

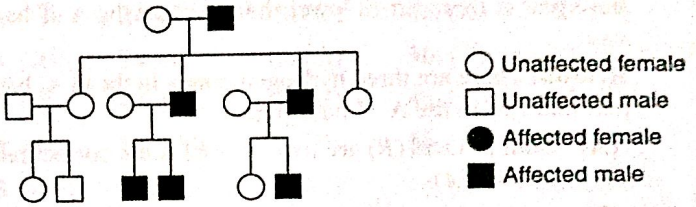
(GATE 2017; 1 Mark)

84. At the transcription start site of a gene, any of the four nucleotides can occur with equal probability p. The Shannon Entropy S, given by $S = -\sum_{i=1}^{4} p_i \ln p_i$, for this start site is_____.

Given data ln (2) = 0.69

(GATE 2017; 1 Mark)

85. A pedigree of an inheritable disease is shown below



This inheritance disease is

(A) X-linked dominant.
(B) X-linked recessive or Y-linked.
(C) only Y-linked.
(D) only X-linked recessive.

(GATE 2017; 2 Marks)

86. A DNA strand of length 25 nm wraps diametrically around the circumference of a spherical histone-octamer once. The radius (nm) of the histone-octamer is _____.

Given data: $\pi = 3.14$

(GATE 2017; 2 Marks)

87. The repeat sequence of telomere in humans is

(A) 5′–TATAAT–3′
(B) 5′–TTAGGG–3′
(C) 5′–GGGCCC–3′
(D) 5′–AAAAAA–3′

(GATE 2018; 1 Mark)

88. If a segment of a sense strand of DNA is 5′-ATGGACCAGA-3′, then the resulting RNA sequence after transcription is

(A) 5′–AGACCAGGTA–3′
(B) 5′–UCUGGUCCAU–3′
(C) 5′–UACCUGGUCU–3′
(D) 5′–AUGGACCAGA–3′

(GATE 2018; 1 Mark)

89. Match the classes of RNA molecules in Group I with their functions in Group II.

Group I	Group II
P. snoRNA	1. Protects germline from transposable elements
Q. piRNA	2. Blocks translation of selected mRNA
R. miRNA	3. Template for telomere elongation
S. snRNA	4. Modification and processing of rRNA
	5. Splicing of RNA transcripts

(A) P–3; Q–5; R–2; S–4 **(B)** P–1; Q–3; R–2; S–5
(C) P–1; Q–4; R–5; S–2 **(D)** P–4; Q–1; R–2; S–5

(GATE 2018; 2 Marks)

90. Determine the correctness or otherwise of the following Assertion (*A*) and the Reason (*R*).

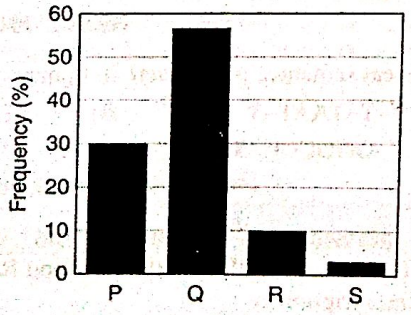
Assertion: The association constant in water for the G–C base pair is three times lower than that for the A–T base pair.

Reason: There are three hydrogen bonds in the G–C base pair and two in the A–T base pair.

(A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(B) (*A*) is false, but (*R*) is true.
(C) Both (*A*) and (*R*) are false.
(D) Both (*A*) and (*R*) are true and (*R*) is not the correct reason for (*A*).

(GATE 2018; 2 Marks)

91. An analysis of DNA-protein interactions was carried out using all DNA-protein complexes in the protein data bank (PDB). The frequency distribution of four amino acid residues, represented as P, Q, R and S, occurring in non-covalent interactions between the protein and DNA backbone is shown below.

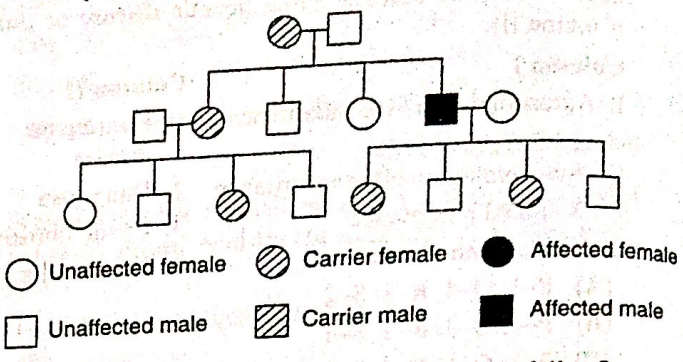


Which one of the following is correct?

(A) P–Lys, Q–Arg, R–Gln, S–Glu
(B) P–Gln, Q–Glu, R–Lys, S–Arg
(C) P–Asn, Q–Asp, R–Arg, S–Lys
(D) P–His, Q–Glu, R–Gln, S–Lys

(GATE 2018; 2 Marks)

92. A pedigree of an inheritable disease is shown below.



What type of inheritance does the disease follow?

(A) Autosomal dominant
(B) X-linked dominant
(C) X-linked recessive
(D) Autosomal recessive

(GATE 2018; 2 Marks)

93. 5′ capping of mRNA transcripts in eukaryotes involves, the following events:

P. Addition of GMP on the 5′ end.
Q. Removal of γ-phosphate of the triphosphate on first base at the 5′ end.
R. 5′-5′ linkage between GMP and the first base at 5′ end.
S. Addition of methyl group to N7 position of guanine.

Which one of the following is the correct sequence of events?

(A) P, Q, R, S
(B) P, R, Q, S
(C) Q, P, R, S
(D) Q, P, S, R

(GATE 2018; 2 Marks)

94. A mutation in a gene that codes for a polypeptide results in a variant polypeptide that lacks the last three amino acids. What type of mutation is this?

(A) Synonymous mutation **(B)** Nonsense mutation
(C) Missense mutation **(D)** Silent mutation

(GATE 2019; 1 Mark)

95. Which of the following processes can increase genetic diversity of bacteria in nature?

P. Conjugation Q. Transformation
R. Transduction S. Transfection

(A) P only **(B)** P and Q only
(C) P, Q and R only **(D)** P, Q, R and S

(GATE 2019; 1 Mark)

96. Which one of the following is coded by the ABO blood group locus in the human genome?

(A) Acyl transferase (B) Galactosyltransferase
(C) Transposase (D) β-Galactosidase

(GATE 2019; 2 Marks)

97. Which one of the following covalent linkages exists between 7-Methyl guanosine (m^{7}G) and mRNAs?

(A) 2′–3′ triphosphate (B) 3′–5′ triphosphate
(C) 5′–5′ triphosphate (D) 2′–5′ triphosphate

(GATE 2019; 2 Marks)

98. In pea plants, purple color of flowers is determined by the dominant allele while white color is determined by the recessive allele. A genetic cross between two purple flower-bearing plants results in an offspring with white flowers. The probability that the third offspring from these parents will have purple flowers is _____ (rounded off to 2 decimal places).

(GATE 2019; 2 Marks)

ANSWER KEY

1. (A)	2. (B)	3. (B)	4. (C)	5. (C)	6. (B)	7. (10^5; 10^4)		8. (D)	9. (B)
10. (C)	11. (A)	12. (Glycosylation)		13. (B)	14. (A), (D)		15. (A)	16. (B)	17. (B)
18. (B)	19. (B)	20. (D)	21. (C)	22. (B)	23. (B)	24. (C)	25. (C)	26. (D)	27. (D)
28. (A)	29. (A)	30. (B)	31. (A)	32. (C)	33. (A)	34. (B)	35. (A)	36. (A)	37. (B)
38. (C)	39. (C)	40. (C)	41. (C)	42. (B)	43. (B)	44. (D)	45. (B)	46. (A)	47. (B)
48. (D)	49. (B)	50. (D)	51. (C)	52. (C)	53. (B)	54. (B)	55. (C)	56. (A)	57. (A)
58. (45%)	59. (A)	60. (A)	61. (C)	62. (A)	63. (0.67)	64. (C)	65. (C)	66. (B)	67. (D)
68. (C)	69. (C)	70. (0.25)	71. (31.25%)		72. (C)	73. (D)	74. (D)	75. (D)	76. (C)
77. (B)	78. (C)	79. (1.56%)	80. (B)	81. (D)	82. (B)	83. (100 min)		84. (1.386)	85. (C)
86. (3.98 nm)		87. (B)	88. (D)	89. (D)	90. (B)	91. (A)	92. (C)	93. (C)	94. (B)
95. (C)	96. (B)	97. (C)	98. (0.75)						

ANSWERS WITH EXPLANATIONS

1. Topic: Molecular Structure of Genes and Chromosomes

Given, G + C content of bacteriophage M13 double stranded DNA is 68%.

Thus, A + T content of bacteriophage M13 double stranded DNA is 32%.

The mRNA formed will be complimentary to the DNA strand. Replace the C with G, G with C, A with U, T with A. Thus, there is no difference in the G and C content in the mRNA and the coding strand. Hence if the DNA (Template strand or Coding strand) has 68% of G + C content then the mRNA, should have 68% of the G + C content.

Answer (A)

2. Topic: Nucleic Acid Replication

DNA helicases are enzymes capable of unwinding duplex DNA to provide the single-stranded DNA templates that are required in processes such as replication and recombination. All helicases separate the strands of a double helix using the energy derived from nucleotide hydrolysis.

Answer (B)

3. Topic: Transcription

Rho-independent termination is a mechanism in prokaryotes that causes RNA transcription to stop and release the newly made RNA. DNA of such mRNA contains specific sites at which transcription is terminated. They share two common features:

- A series of 4 to 10 consecutive base pairs, with the A's on the template strand. The transcribed RNA is terminated in or just past this sequence.
- A G + C rich region with a palindromic sequence that immediately precedes the series of A - T's.

The RNA transcript of this region, called an intrinsic terminator, can therefore form a self-complementary "hairpin" structure that is terminated by several U residues. The structural stability of an RNA transcript at its terminator's G + C-rich hairpin and the weak base pairing of its oligo(U) tail to template DNA appear to be important factors in ensuring proper chain termination. The formation of the G + C-rich hairpin causes RNA Polymerase (RNAP) to pause for several seconds at the termination site. This probably induces a conformational

change in the RNAP that permits the non-template DNA strand to displace the weakly bound oligo(U) tail from the template strand, thereby spontaneously terminating transcription.

Answer (B)

4. Topic: Transcription

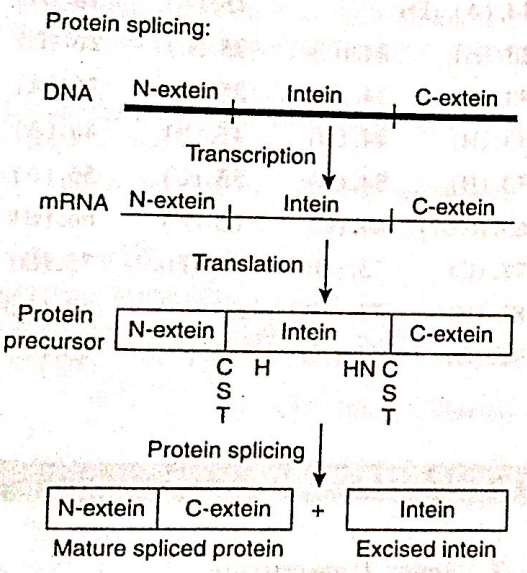
Eukaryotic nuclei contain three distinct types of RNA polymerase that differ in the RNAs they synthesize:

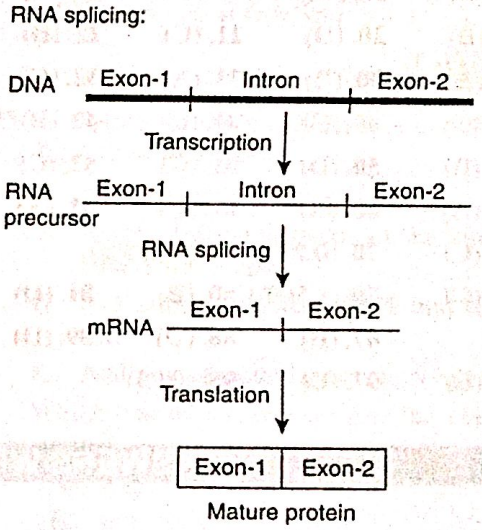
- RNA polymerase I (RNAP I), which is located in the nucleoli, synthesizes the precursors of most rRNAs.
- RNA polymerase II (RNAP II), which occurs in the nucleoplasm, synthesizes the mRNA precursors.
- RNA polymerase III (RNAP III), which also occurs in the nucleoplasm, synthesizes the precursors of 5S rRNA, the tRNAs, and a variety of other small nuclear and cytosolic RNAs.

Answer (C)

5. Topic: Translation

Protein splicing is an intramolecular reaction of a particular protein in which an internal protein segment (called an intein) is removed from a precursor protein with a ligation of C-terminal and N-terminal external proteins (called exteins) on both sides. The splicing junction of the precursor protein is mainly a cysteine or a serine, which are amino acids containing a nucleophilic side chain. The protein splicing reactions which are known now do not require exogenous cofactors or energy sources such as adenosine triphosphate (ATP) or guanosine triphosphate (GTP). Normally, splicing is associated only with pre-mRNA splicing.





Answer (C)

6. Topic: Molecular Basis of Genetic Diseases

Phenylketonuria (PKU) is a genetic metabolic disorder that increases the body's levels of phenylalanine. Phenylalanine is one of the building blocks (amino acids) of proteins. Humans cannot make phenylalanine, but it is a natural part of the foods we eat. However, people do not need all the phenylalanine they eat, so the body converts extra phenylalanine to another harmless amino acid, tyrosine. People with PKU are not able to produce phenylalanine hydroxylase, thus cannot break down the extra phenylalanine to convert it to tyrosine. This means phenylalanine builds up in the person's blood, urine, and body. If PKU is not treated, phenylalanine can build up to harmful levels in the body.

Answer (B)

7. Topic: Molecular Structures of Genes and Chromosomes

(i) Given, length of double stranded DNA is 34 μm, i.e., 3.4×10^5 Å. We know that each turn in helix rises at 34 Å. To find the number of base pairs in DNA molecule, divide total length of DNA by length of a turn of DNA and multiply the result by ten as a turn consist of ten base pairs.

$$\frac{3.4\times10^5}{34}\times10 = 10^5$$

Thus, there are 10^5 base pairs in the DNA.

(ii) Similarly, number of turns can be calculated by dividing total length of DNA by length of a turn of DNA.

$$\frac{3.4\times10^5}{34} = 10^4$$

Thus, there are 10^4 turns in DNA.

Answer (10^5; 10^4)

8. Topic: Microbial Genetics

Heterologous protein expression is the introduction of either complementary DNA (cDNA) or RNA (cRNA) encoding for a protein of interest from one species into the cell of another species, such that the hosts' cellular machinery expresses the foreign protein. Major challenges faced by this expression system are the expression of unusually difficult/complex proteins with rare codons or membrane and toxic proteins. The production level of heterologous proteins greatly depends on the characteristics of the host cell, number of plasmids carrying the recombinant protein to be expressed, the promoter used and most importantly on the composition of the medium, showing that production can be limited at any level. Choosing the correct expression system/protocols are obligatory for the substantial expression of protein in the native form. Large number of plasmids, their compatible hosts and culture conditions can be used to express recombinant proteins in large amounts and in native form.

Answer (D)

9. Topic: Translation

Characteristics of the genetic code:

- Triplet nature: A triplet code could make a genetic code for 64 different combinations (4 × 4 × 4) genetic code and provide plenty of information in the DNA molecule to specify the placement of all 20 amino acids.
- Universality: Same genetic code is found valid for all organisms ranging from bacteria to man. It means that the same sequences of 3 bases encode the same amino acids in all life forms from simple microorganisms to complex, multi-celled organisms such as human beings.
- Degeneracy/Redundancy: The triplet code allows many amino acids to be specified by more than one codon. Such a code is said to be degenerate. Because of the degeneracy of the genetic code, a point mutation in the third position of a codon seldom alters the specified amino acid.

Answer (B)

10. Topic: Molecular Structure of Genes and Chromosomes

The major groove is wider than the minor groove in DNA, and many sequence specific proteins interact in the major groove. The N7 and C6 groups of purines and the C4 and C5 groups of pyrimidines face into the major groove, thus they can make specific contacts with amino acids in DNA-binding proteins. Thus, specific amino acids serve as H-bond donors and acceptors to form H-bonds with specific nucleotides in the DNA.

Answer (C)

11. Topic: Nucleic Acid Replication, Transcription, Translation

"Regular" genetics begins, with a mutant phenotype, proceeds to show the existence of the relevant gene responsible for it. However, reverse genetics, works in the opposite direction.

Genetics	Reverse genetics
Mutant phenotype	Protein sequence
↓	↓
Mutant allele	DNA sequence
↓	↓
DNA sequence	Mutant allele
↓	↓
Protein sequence	Mutant phenotype

Often the starting point of reverse genetics is a protein or a gene "in search of a function," or, looked at another way, a protein or a gene "in search of a phenotype." For example, a cloned wild-type gene detected by sequencing as an open reading frame (ORF) of unknown function can be subjected to some form of *in vitro* mutagenesis and reinserted back into the organism to determine the phenotypic outcome.

Answer (A)

12. Topic: Translation

Glycosylation is a major post-translational modification in eukaryotes. It is responsible for the formation of cellular glycans which are often attached to proteins and lipids. Glycosyltransferase and glycosidases are enzymes responsible for glycosylation of many proteins. Glycoproteins, which are commonly distributed in eukaryotic cells, are rarely presented in prokaryotic organisms because cellular organelles essential for glycosylation are missing in these organisms.

Answer (Glycosylation)

13. Topic: Molecular Structure of Genes and Chromosomes

RNA, is a stable molecule as the presence of negative charge on sugar-phosphate backbone protects it from attack by hydroxyl ions (–OH) that could lead to hydrolytic cleavage. But, the presence of 2′-hydroxyl (–OH) group makes the RNA susceptible to base-catalyzed hydrolysis. Moreover, RNA is also prone to auto-hydrolysis when it is single stranded. This spontaneous cleavage reaction takes place in basic solutions, where free hydroxyl ions in solution can easily deprotonate the 2′-hydroxyl (–OH) group of the ribose sugar. However, if this 2′-hydroxyl (–OH) group is removed from the ribose sugar then the rate of such base- catalyzed hydrolysis are decreased by approximately 100 folds under extreme conditions.

Answer (B)

14. Topic: Translation

Bacterial protein expression systems are popular because bacteria are easy to culture, grow fast and produce

high yields of recombinant protein. However, multi-domain eukaryotic proteins expressed in bacteria often are non-functional because the cells are not equipped to accomplish the required post-translational modifications or molecular folding. Also, many proteins become insoluble as inclusion bodies that because of the activity of enzymes produced by host bacteria.

Answer (A), (D)

15. Topic: Microbial Genetics

On introduction of viral genes in the plant genome in fully functional state, the genes get integrated in the plant chromosomes and become a part of it. Thus, during replication or transcription of DNA, that viral part gets replicated or transcribed like the rest of the genome and thus exhibit Mendelian inheritance.

Answer (A)

16. Topic: Regulatory Mechanisms in Eukaryotes

RNA interference is a type of post-transcriptional gene silencing. Short interfering RNAs (siRNAs) or microRNAs (miRNAs), depending on their origin, down-regulate gene expression by binding to complementary mRNA molecules. An siRNA is transferred to a 250 to 500 kD enzyme complex known as the RNA-induced silencing complex (RISC). RISC has at least four protein components, one of which is an ATP-dependent RNA helicase that separates the two strands of the siRNA. The strand whose 5′ end has the lower free energy of binding, the guide RNA, is bound by the RISC, whereas its complementary strand, the passenger RNA, is cleaved and discarded.

Answer (B)

17. Topic: Transposable Elements

Around 42% of the human genome consists of three types of retrotransposons:

- Long interspersed nuclear elements (LINEs) are 6- to 8 kb-long segments. The great majority of these molecular parasites have accumulated mutations that render them transcriptionally inactive, but a few still appear capable of further transposition.
- Short interspersed nuclear elements (SINEs) consist of 100 to 400 bp elements. The most common SINEs in the human genome are members of the *Alu* family.
- Retrotransposons with long terminal repeats (LTRs).

Answer (B)

18. Topic: Population Genetics

- Orthologous genes are homologous genes that diverged after evolution gives rise to different species, an event known as speciation. The genes generally maintain a similar function to that of the ancestral gene that they evolved from.
- Paralogous genes are homologous genes that have diverged within one species. Unlike orthologous genes, a paralogous gene is a new gene that holds a new function. These genes arise during gene duplication where one copy of the gene reverts to a mutation that gives rise to a new gene with a new function, though the function is often related to that role of the ancestral gene.
- A proteome is the complete set of proteins expressed by an organism. It is an expression of an organism's genome.
- A transgene is an exogenous gene that has been introduced into the genome of another organism, and a transgenic species is one whose genome has been genetically altered.

Answer (B)

19. Topic: Regulatory Mechanisms in Eukaryotes

In the process of gene silencing in eukaryotes, the trigger double-stranded RNA is chopped up into ~21- to 23-nt fragments (that is, siRNAs), each of whose strands has a 5′ phosphate and a 2-nt overhang at its 3′ end. The cleavage reaction is catalyzed by an ATP-dependent RNase named Dicer, a homodimer of ~1900-residue subunits in animals that is a member of the RNase III family of double-strand-specific RNA endonucleases.

Answer (B)

20. Topic: Regulatory Mechanisms in Prokaryotes and Eukaryotes

Apoptosis involves a family of proteases known as caspases (for cysteinyl aspartate-specific proteases), which are cysteine proteases whose mechanism resembles that of serine proteases, but with Cys replacing the active site Ser. Caspases cleave target polypeptides after an Asp residue.

Answer (D)

21. Topic: Microbial Genetics

The Ames test, devised by the American microbiologist Bruce Ames, is used to test whether substances induce mutations in certain strains of *Salmonella* (auxotrophs) that have lost their ability to synthesize the amino acid histidine. These strains easily undergo another mutation that restores their ability to synthesize histidine. The Ames test is based on the hypothesis that if a substance is a mutagen, it will increase the rate at which these organisms revert to being histidine synthesizers. Furthermore, the more powerful a substance's mutagenic capacity, the greater the number of reverted organisms it causes to appear. In practice, the organism is grown in the presence of a test substance. If any organisms regain the ability to synthesize histidine, the substance is suspected of being a mutagen, and therefore may act as a carcinogen, because cancer is often linked to mutation.

Answer (C)

22. Topic: Translation

The peptidyl transferase center that catalyzes peptide bond formation is located on the large subunit and consists entirely of rRNA. The X-ray structures of the large ribosomal subunit in complex with aminoacyl-tRNA and peptidyl-tRNA indicate that the ribosome uses an induced fit mechanism, as occurs in enzymes. Conformational changes in the 23S rRNA, presumably triggered by proper binding of the aminoacyl-tRNA in the A site, orient the ester group of the peptidyl-tRNA for nucleophilic attack.

Answer (B)

23. Topic: Mutations and Mutagenesis

Transgenics describes the process of altering host organism's genome. Biotechnology has tried to develop transgenics through gene engineering. This generally involves two strategies:

- Gene addition: In this, genotype (genetic constitution) is altered by providing it with one or more new genes. The foreign DNA, or "transgene," that is transferred to the recipient can be from other individuals of the same species or even from unrelated species.
- Gene subtraction: In this, one or more of the already existing genes are inactivated by employing various genetic engineering methods.

It is done to determine an unknown gene's function; to analyze the malfunction of a mutated gene; to model human disease; and to provide better agricultural and pharmaceutical products by making transgenic plants and animals.

Answer (B)

24. Topic: Transcription

During transcription and DNA replication, the DNA helix can become either over-wound or under-wound. For instance, movement of the replication fork generates positive supercoils in the unreplicated portion of the DNA ahead of the fork. Such tensional problems also exist when transcribing DNA to make an RNA copy for protein synthesis. The mechanical strain that builds up during these processes is relieved by DNA gyrase that moves along the DNA ahead of the replication fork and removes positive supercoils by introducing negative supercoil. This is done by cleaving both strands of the DNA duplex by enzymes called topoisomerases passing a segment of DNA through the double-stranded break to the other side, and then sealing the cuts. This process requires the energy released during ATP hydrolysis.

Answer (C)

25. Topic: DNA Replication

Terminal deoxynucleotidyl Transferase (TdT) is a unique enzyme. It is a template-independent DNA polymerase that uses only single-stranded DNA as nucleic acid substrate. It catalyzes the repetitive addition of deoxyribonucleotides to the 3′-OH of oligodeoxyribonucleotides and single-stranded and double-stranded DNA. It requires an oligonucleotide of at least three nucleotides to serve as a primer. With RNA as template TdT shows variable performance, which strongly depends upon the tertiary structure of acceptor RNA 3′-end and the nature of nucleotide. Generally, it is lower than using DNA as a template.

Answer (C)

26. Topic: Molecular Structure of Genes and Chromosomes

In eukaryotes, DNA is organized into chromatin, which is basically inaccessible to proteins regulating DNA-templated processes such as gene transcription. The fundamental unit of chromatin is the nucleosome, which consists of 147 bp of DNA wrapped 1.7 times around a core histone octamer in a left-handed superhelix. Nucleosome free regions of eukaryotic chromosome DNA are specifically vulnerable to nuclease attack perhaps because they are not wrapped in histone as nucleosomes. This increases the availability of the DNA to binding by proteins, such as transcription factors and nuclease. These are known as nuclease-hypersensitive sites.

Answer (D)

27. Topic: Translation

- Transglycosylation is a mechanism for glyosidic bond formation, particularly during polysaccharide synthesis.
- Autoglycosylation is the reaction in which a carbohydrate, i.e., a glycosyl donor, is attached to a hydroxyl or other functional group of another molecule (a glycosyl acceptor).
- Autopeptidation reactions are elongation reactions of the ribosomal or multienzyme system.
- Transpeptidation is a reaction involving the transfer of one or more amino acids from one peptide chain to another, by transpeptidase action, or within a peptide chain itself, as in bacterial cell wall synthesis.

Answer (D)

28. Topic: Extra Chromosomal Inheritance

Cytoplasmic male sterility (cms), a condition under which a plant is unable to produce functional pollen, anther of male gamete, although the potential for female reproduction remains. It is widespread among higher plants. It occurs due to some specific nuclear and mitochondrial interactions due to defect in mitochondria. It is a maternally inherited trait that facilitates hybrid seed production of many crops and allows breeders to harness yield gains associated with hybrid vigor.

In many cases, it has been found that male fertility can be restored by nuclear-encoded fertility restorer (*Rf*) gene(s). The key role in the nuclear genetic control of cms is played by *Rf* genes, which are responsible for the restoration of male fertility upon interaction with cms-inducing cytoplasm. These genes, which encode pentatricopeptide repeat proteins, affect the expression of the cms gene post-transcriptionally. This results in the disappearance of the cms protein, at least in the male organs.

Answer (A)

29. Topic: Mendelian Inheritance

Curved wings: *CC*; Short wings: *cc*

Tan body: *TT*; Dark body: *tt*

On crossing true breeding curved wings and dark bodies with true breeding short wings and tan body:

P *CCTT* × *cctt*

F_1 *CcTt* (Curved tan)

On its self-cross we will obtain:

	CT	*Ct*	*cT*	*ct*
CT	*CCTT* Curved tan	*CCTt* Curved tan	*CcTT* Curved tan	*CcTt* Curved tan
Ct	*CCTt* Curved tan	*CCtt* Curved dark	*CcTt* Curved tan	*Cctt* Curved dark
cT	*CcTT* Curved tan	*CcTt* Curved tan	*ccTT* Short tan	*ccTt* Short tan
ct	*CcTt* Curved tan	*Cctt* Curved dark	*ccTt* Short tan	*cctt* Short dark

F_2 offspring: 45 Curved tan: 15 Short tan: 16 Curved dark long: 6 Short dark.

- The F_1 offspring are all curved tan. Thus, curved wing is dominant over short wing and tan body is dominant over dark body.
- The F_2 generation is segregated to an approximate ratio of 9:3:3:1 and there is a double recessive offspring group short dark. Thus, it is showing Mendelian pattern of inheritance.

Answer (A)

30. Topic: Chromosomal Variation

- Polyploidy refers to change in number of sets of chromosomes. Such as tetraploid contain four homologous set of chromosomes.
- Euploid is a condition in which cell contains chromosome number that is an exact multiple of haploid cell content.
- Aneuploidy refers to the presence of unusual number of chromosomes in a cell such as aneuploidy of sex chromosomes: Turner's syndrome {XO; $2n - 1$ or 45 (44A + X)}, Klinefelter's syndrome (XXY or XXXY or $2n + 1$); aneuploidy of autosomes: Down's syndrome (trisomy of 21st chromosome), Edward's syndrome (trisomy of 18th chromosome).

Answer (B)

31. Topic: Microbial Genetics

Molecular chaperons are the class of proteins that assist in covalent folding of the newly synthesized polypeptides as well as assembly of subunits. Thus, it helps in attaining the functionality of the proteins. In addition, it prevents the assembly of subunits into non-functional structure.

Answer (A)

32. Topic: Translation

Ubiquitation refers to the addition of ubiquitin group to the protein. It affects proteins in many ways: it can mark them for degradation via the proteasome, alter their cellular location, affect their activity, and promote or prevent protein interactions.

Answer (C)

33. Topic: Mutations and Mutagenesis

Methyl-nitro-nitrosoguanidine (NTG) is an effective chemical mutagen and carcinogen. It acts by adding alkyl groups to the O6 of guanine and O4 of thymine, which can lead to transition mutations between GC and AT, and there is no transversion occurs. Transitions are interchanges of purines (A G) or of pyrimidines (C T): they therefore involve bases of similar shape.

Answer (A)

34. Topic: Nucleic Acid Replication

Enhancer is a short (50–1500 bp) region of DNA that are *cis*-acting sequence and can be bound by proteins (activators/transcription factors) to escalate the transcription of a particular gene, and not replication.

Answer (B)

35. Topic: Transcription and Translation

- The Shine-Dalgarno (SD) sequence is a site for ribosome binding in bacterial messenger RNA and is located around 8 bases upstream of the start codon AUG, i.e., AGGAGGU. It helps to initiate protein synthesis (translation) by aligning the ribosome with the start codon.
- Leucine zippers are a dimerization domain of the bZIP (Basic-region leucine zipper) class of eukaryotic transcription factors.
- An aminoacyl-tRNA synthetase (aaRS or ARS), also called tRNA-ligase, is an enzyme that attaches the correct amino acid onto its tRNA.

- Gene silencing is described as the "switching off" of a gene by a mechanism other than genetic modification, e.g., RNA interference (RNAi) use the gene's own DNA sequence of gene to turn off the cellular mechanism.

Answer (A)

36. Topic: Microbial Genetics

The lac (lactose) operon of *Escherichia coli* is a group of three structural genes (*z*, *y* and *a*) encoding proteins involved in lactose metabolism. A gratuitous inducer is a molecule which is analog to another molecule that induces transcription for a specific product, and which can also initiate transcription for that product when the official inducer is absent, such as IPTG (isopropyl thiogalactoside) for the Lactose operon.

Answer (A)

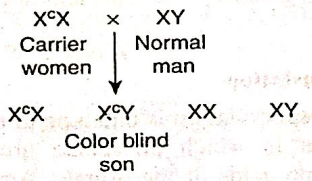
37. Topic: Nucleic Acid Replication

The DNA polymerase III has 5′→3′ polymerase activity as well as has proofreading capabilities that correct replication mistakes by means of exonuclease activity working in 3′→ 5′. AUG (methionine) is usually used as the starting codon, and essential to begin proteins synthesis.

Answer (B)

38. Topic: Molecular Basis of Genetic Diseases

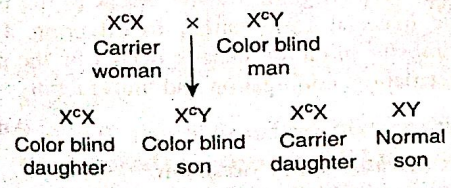
25% will be the probability of having the color-blind son to a woman with phenotypically normal parents and a color-blind brother, and is married to a normal man.



Answer (C)

39. Topic: Molecular Basis of Genetic Diseases

25% will be the probability of having the color-blind daughter to a phenotypically normal woman, who already had one color-blind son, and is married with a color-blind man.



Answer (C)

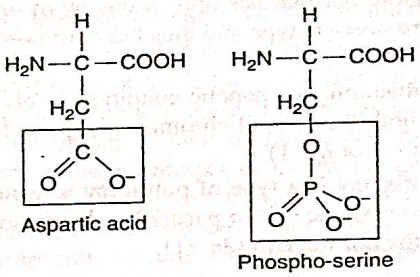
40. Topic: DNA Damage and Repair

DNA methylation is used by the cell to distinguish the parental DNA strand from newly synthesized strand. Cysteine or adenine residues are methylated periodically throughout the genome in different organisms e.g., in *E. coli* adenine residue of GATC sequence is methylated with the help of DNA adenine methyltransferase (*Dam*). It creates hemimethylated stage after replication for few seconds during which mismatch repair system of the cell checks for errors in newly synthesized DNA strand.

Answer (C)

41. Topic: Regulatory Mechanisms in Prokaryotes and Eukaryotes

In regulation, phosphorylation is a very common method used for activating or deactivating the protein. Most commonly phosphorylated residues in proteins are serine, tyrosine, and threonine. However, there are amino acids, which can mimic phosphorylated proteins and are called phosphomimetics, e.g., aspartic acid mimics phosphorylated serine.



Answer (C)

42. Topic: Mutations and Mutagenesis

Nonsense mutation is a point mutation, which leads to generation of stop codon in transcript mRNA. It leads to premature or truncated polypeptide synthesis. It is advised to introduce another mutation in transfer RNA gene in order to obtain full-length polypeptide. Mutated tRNA may read the stop codon as coding codon and continues the polypeptide synthesis to its end.

Answer (B)

43. Topic: Extra Chromosomal Inheritance

The direction of shell coiling in the snail *Limnaea peregra* is a classic example of maternal inheritance, i.e., inheritance caused by mitochondrial DNA. Usually, mitochondria are passed from egg rather than sperm.

Answer (B)

44. Topic: Molecular Structure of Genes and Chromosomes

Synteny refers to the distribution pattern of genes on chromosome. It is used to check the conservative order

of genes on chromosomes of different species which is known as shared synteny.

Answer (D)

45. Topic: Complementation

Beginning with the initial sequence 7 bp sequence
5′ ATGAGCC 3′
The complement created by base pairing is:
3′ TACTCGG 5′
Its reverse complement will be
5′ GGCTCAT 3′
Thus, the inverted repeat sequence is
5′ ATGAGCCGGCTCTA 3′

Answer (B)

46. Topic: Chromosomal Variation

- Disomic have an extra chromosome in the haploid state ($n + 1$) that is homologous to an existing chromosome in this set.
- In monosomic, a cell or individual that is basically diploid but that has only one copy of one particular chromosome type and thus has chromosome number $2n - 1$.
- Nullisomic is a genetic condition involving the lack of both the normal chromosomal pairs for a species ($2n - 2$ or $n - 1$).
- A trisomy is a type of polysomy in which there are three instances of a particular chromosome, instead of the normal two ($2n + 1$).

Answer (A)

47. Topic: Molecular Basis of Genetic Diseases

Given,

Probability of getting the disease in a child $= \frac{1}{4}$

Probability of not getting the disease in a child $= \frac{3}{4}$

The probability that exactly one sibling is affected by this disease $= {}^2C_1 \frac{3}{4} \times \frac{1}{4} = 2 \times \frac{3}{16} = \frac{3}{8}$

Answer (B)

48. Topic: Microbial Genetics

Given: Volume, $V = 10^{-12}\ cm^3 = 10^{-12} \times 10^{-6}\ m^3 = 10^{-18}\ m^3$
$= 10^{-18} \times 10^3\ L = 10^{-15}\ L$

Number of molecules = 60

$$\text{Molar concentration} = \frac{\text{No. of molecules}}{N_A \times \text{Volume(L)}}$$

$$= \frac{60}{6.022 \times 10^{23} \times 15} = 100\ nM$$

Answer (D)

49. Topic: Microbial Genetics

The repressor has a binding affinity (K_d) of 10^{-8} M with lactose and molar concertation of repressor in the cell is 100 nM or 10^{-7} M. Thus, they will bind to the repressor region and operon will be repressed in presence of lactose.

Answer (B)

50. Topic: Molecular Structure of Genes and Chromosomes

DNA contains deoxyribose sugars, whereas RNA contains ribose sugars, i.e., few extra OH group at 2nd position. As we know, OH is reactive species and hence RNA is less stable. In addition, under alkaline conditions, presence of extra OH group in ribose sugar makes it (phosphate group) more susceptible to nucleophilic attack which leads to cleavage of phosphor-di-ester bond in RNA.

Answer (D)

51. Topic: Nucleic Acid Replication, Transcription, Translation

- Methylation refers to the addition of methyl group to any substance. It is the modification common to both DNA and protein. Generally, cytosine and adenine are usually methylated in DNA. It helps in regulating gene expression in prokaryotes and eukaryotes. In addition, in *E. coli* DNA methylation also helps in differentiating the newly synthesized and parent DNA after replication.
- In proteins, methyl transferase generally adds methyl group to the nitrogen side chains of lysine and arginine residues. It is post-translational modification which regulates protein-protein interaction.

Answer (C)

52. Topic: Translation

Protein phosphorylation is one type of post-translational modification in which phosphate group is added to certain amino acids of the protein. Amino acids which are most commonly phosphorylated are serine, threonine, and tyrosine.

Answer (C)

53. Topic: Horizontal Gene Transfer

Horizontal gene transfer refers to the transfer of genetic material between unicellular or multi-cellular cells. In contrast, vertical gene transfer refers to the transfer of genetic material from parent to offspring. Horizontal gene transfer in nature usually occurs by the process of transformation, conjugation and transduction.

Answer (B)

54. Topic: Chromosomal Variation

Allotriploid individuals refer to the hybrid individuals with three sets of chromosomes derived from two different

ancestral species. They derive $1n$ from one species and $2n$ from other. So, here number of chromosomes in allotriploid hybrid can be calculated as 18 + 28 or 36 + 14 which is equal to 46 or 50.

Answer (B)

55. Topic: Nucleic Acid Replication

- DNA gyrase is an enzyme of class topoisomerase which releases the strain of supercoiling taking place at the time of unwinding of DNA.
- Primase is a DNA dependent RNA polymerase which adds RNA primers to the lagging strand of DNA. It provides flanking 3′–OH for DNA polymerase.
- DNA polymerase is a DNA dependent DNA polymerase enzyme. There are five DNA polymerases in bacteria, namely DNA pol I to V. DNA pol I is involved in excision repairs and RNA primers removal as it has both 3′– 5′ and 5′ – 3′ exonuclease activity.
- DNA pol II is involved DNA repair as it has 3′ – 5′ exonuclease activity.
- DNA pol III holoenzyme is the primary enzyme involved actively in DNA replication as it has high processivity and least error rate comparatively.
- DNA pol IV and V are usually involved in SOS response and translation synthesis.

Answer (C)

56. Topic: Molecular Basics of Genetic Diseases

The cross between normal male and heterozygous female can be drawn as:

Parents: XY (normal male) × X^hX (affected female)

Gametes: X, Y and X^h X

Cross:

Gametes	X	Y
X^h	X^hX	X^hY
X	XX	XY

Hence, half of the sons and half of the daughters are diseased.

Answer (A)

57. Topic: Nucleic Acid replication, Transcription and Translation

- RNase P is a type of ribonuclease (ribozyme). It is responsible for generating the mature 5′ end of tRNAs by a single endonucleolytic cleavage of their precursors.
- RNase H is an endonuclease that specifically cleaves the phosphodiester bonds of RNA hybridized to DNA.
- A small nuclear RNA (snRNA) is one of many small RNA species confined to the nucleus. They are involved in splicing or other RNA processing reactions.
- In nuclear polyadenylation, a poly(A) tail is added at the end of RNA during transcription. CstF (Cleavage stimulation factor) assembles the protein complex on the 3′ end to promote the synthesis of a polyadenine tail, which results in a mature mRNA molecule which is exported from the cell nucleus to the cytosol for translation.

Answer (A)

58. Topic: Population Genetics

Given,

Number of people with allele for polydactyly = 80

Number of people who show polydactyly = 36

Extent of penetrance of allele in the population can be calculated as:

$$\frac{36}{80} \times 100 = 45\%$$

Answer (45%)

59. Topic: Molecular Structure of Genes and Chromosomes

Two grooves of different width are formed between adjacent turns of the helix — a wider major groove and a narrower minor groove. The major groove is wider than the minor groove in DNA, and many sequence specific proteins interact in the major groove. The N7 and C6 groups of purines and the C4 and C5 groups of pyrimidines face into the major groove, thus they can make specific contacts with amino acids in DNA-binding proteins. Thus, specific amino acids serve as H-bond donors and acceptors to form H-bonds with specific nucleotides in the DNA.

Answer (A)

60. Topic: Molecular Structure of Genes and Chromosomes

(i) Messenger RNA (mRNA) serves as a template to direct protein synthesis. It is synthesized from DNA by the process of transcription in 5′→3′ direction. The growing mRNA molecule has a 5′ triphosphate group.

(ii) Transfer RNA (tRNA) is a set of small, compact molecules that function to deliver amino acids to the ribosomes for assembly into proteins. They function as adaptors between amino acids and the codons in mRNA during translation (or protein synthesis). The tRNA has clover leaf secondary structure. Starting from the 5′ end, they have the following common features:

- A 5′ terminal monophosphate group.
- The amino acid acceptor end where the amino acid residue carried by the tRNA is covalently attached to its 3′ terminal OH group.

- It is the DNA unit that frequently contains the modified base dihydrouridine (D).
- Anticodon loop that contains the anticodon.
- TψC or T arm that usually contains the sequence TψC (where ψ is the symbol for pseudouridine).
- A 3′ CCA sequence with a free OH group.

Answer (A)

61. Topic: Population Genetics

The population is finite and closed, thus absence of non-random mating and random genetic drift is negligible. Thus, the most likely reason for change in the frequency of PP genotype is natural selection. Natural selection is a mechanism by which evolution takes place. It is also referred to as the survival of the fittest. This phrase was given by Herbert Spencer. Here, fitness refers to the reproductive fitness of an organism. This implies that the organism is able to survive up to the reproductive stage and is able to bear offspring and hence is able to pass on the genes to the next generation. It is also known as Darwinian fitness. In any given generation, some individuals are always likely to survive more than the others. They show reproductive fitness.

Answer (C)

62. Topic: Molecular Structure of Genes and Chromosomes

Topology theory indicates that the sum of Twist and Writhing number equals to linking number.

$$\therefore \quad L = T + W$$

where, L is the linking number to be calculated.

T is the twist = 191

W is the writhing number = –4

Putting these values in the formula above we get:

$L = 191 - 4 = 187$

Length of circular DNA = 2 kb = 2000 bp

Therefore, number of base pairs/turn = 2000/187 = 10.69

Answer (A)

63. Topic: Population Genetics

Hardy–Weinberg equilibrium, states that the amount of genetic variation in a population will remain constant from one generation to the next if the disturbing factors are absent. It is given by:

$$p^2 + 2pq + q^2 = 1$$

Here, the population given is of 10,000 individuals.

where, p = frequency of the P allele in the population

q = frequency of the p allele in the population

p^2 = frequency of the homozygous genotype

$PP = 2500/10000 = 0.25$

q^2 = frequency of the homozygous genotype pp

$2pq$ = frequency of the heterozygous genotype

$Pp = 1000/10000 = 0.1$

Putting these values in the Hardy–Weinberg equation we get:

$$0.25 + 0.1 + q^2 = 1$$

$$\text{or} \quad q^2 = 1 - 0.35 = 0.65$$

$$\text{or} \quad q = \sqrt{0.65} = 0.87$$

Answer (0.87)

64. Topic: Translation

The mRNA codons in the A and P sites are exposed on the interface side of the 30 S subunit of 70 S ribosome, whereas its 5′ and 3′ ends are bound to ribosomal component of RNA and protein. The mRNA's Shine-Dalgarno sequence, which is located near its 5′ end, is base paired, as expected, with the 16S rRNA's anti-Shine-Dalgarno sequence, which is situated close to the 3′ site. The proteins that in part form a tunnel through which the mRNA enters the ribosome probably function as a helicase to remove secondary structures from the mRNA that would otherwise interfere with tRNA binding.

Answer (C)

65. Topic: Gene Interaction

Clearly from the genetic cross, it is seen that W and X are closer to each other than Y and Y. Simultaneously, X and W are close to each other, while Y is farthest from X. Thus, the relative positioning of the genes would be Y–W–X and y–w–x.

Relative distances of genes with respect to each other are as follows:

Distance of W gene from Y, Y–W = 5 units and W from X, W–X = 11 units.

Answer (C)

66. Topic: Molecular Structure of Genes and Chromosomes

The two main base/sugar conformation in nucleic acids are *syn* and *anti*. The *anti*-conformation of the glycosidic bond typical for DNA allows specific hydrogen bond formation between the base paired bases in DNA double helix. Two preferred conformations of pentose sugar (sugar puckers) are 2′ endo (2′–OH is pointing the same way as the base) and 3′ endo (3′–OH is pointing the same way as the base). Sugar conformation is critical for the nucleic acid structure because it defines how the nucleotides are linked to each other in space. So, the orientation of the nitrogenous bases to the pentose sugars, and the puckering of the sugar, is done in *anti*, and 2′-endo in B form DNA.

Answer (B)

67. Topic: Mendelian Inheritance

Dioecy is characterized by a species having distinct male and female organisms. This is opposed to hermaphroditic species, or more correctly, monoecious species, in which in one individual both male and female reproductive organs are present. Dioecious reproduction is biparental reproduction. The embryo sac, where one fuses with the egg and forms a zygote and the other fuses with the two polar nuclei of the central cell and forms a triple fusion, or endosperm, nucleus. This is called double fertilization because the true fertilization (fusion of a sperm with an egg) is accompanied by another fusion process (that of a sperm with the polar nuclei) that resembles fertilization.

On crossing $XX \times XY$

F_1 XX XY

50% 50%

For endosperm another fertilization with sperm gametes. There is X and Y gametes for sperm which fuses with XX producing 50% XXX and 50% XXY genotype endosperm

Answer (D)

68. Topic: Translation

Tuples are sequences. Number of tuples $= L - k + 1$

where,

L, Total length of sequence $= 13$

K, tuples $= 3$

Number of 3 tuples $= 13 - 3 + 1 = 11$

Start Position	3-Tupule
1	MAD
2	ADC
3	DCM
4	CMW
5	MWD
6	WDI
7	DIS
8	ISE
9	SEA
10	EAS
11	ASE

Answer (C)

69. Topic: Translation

Number of standard amino acid = 20

Number of distinct proteins containing n amino acids $= 20^n$

Number of distinct proteins containing 100 amino acids $= 20^{100}$

Answer (C)

70. Topic: Population Genetics

Frequency of respective genotype is:

$$f(AA) = 0.3$$
$$f(Aa) = 0.4$$
$$f(aa) = 0.3$$

Frequency of allele A in next generation,

$$p = f(AA) + \frac{f(Aa)}{2}$$
$$= 0.3 + \frac{0.4}{2}$$
$$= 0.5$$

Thus, the frequency of genotype AA, $p^2 = 0.5 \times 0.5 = 0.25$

Answer (0.25)

71. Topic: Mendelian Inheritance

On crossing heterozygous tall plant (Tt) with a homozygous dwarf plant (tt).

	T	t
t	Tt	tt
t	Tt	tt

Thus, 50% of the progeny would be tall and 50% would be dwarf.

Probability of progeny with genotype Tt, $p(Tt) = 0.5$

Probability of progeny with genotype tt, $p(tt) = 0.5$

Now, the probability of exactly two of the five seeds to be dwarf can be calculated by using Binomial theorem,

$$P = \frac{n!}{x!y!}[P(Tt)]^x[P(tt)]^y$$

where,

n, total number of events = 5

x, number of times $P(Tt)$ to occur = 3

y = number of times $P(tt)$ to occur = 2

$$P = \frac{5!}{3!2!}[0.5]^3[0.5]^2$$
$$= \frac{5\times4\times3!}{3!2!}[0.5]^5 = 10\left[\frac{1}{2}\right]^5 = \frac{5}{16}$$

$$P\text{ (in \%)} = \frac{5}{16}\times100 = 31.25\%$$

Answer (31.25%)

72. Topic: Translation

In first case, only two codons are possible: UCU, CUC

In second case, three codons are possible: CCU, CUC, UCC

As per the Wobble hypothesis, the unique amino acid codon is only CCU in second case, synthesizing a new amino acid.

Answer (C)

73. Topic: Microbial Genetics

- Mutant strain *a*⁻ having defective *a* gene is not able to develop color with either X, Y or Z intermediate. Thus, product is not being produced in any of the cases. Hence, it must encode for enzyme converting Z to the product.
- Mutant strain *b*⁻ having defective *b* gene is developing color with Y and Z intermediate, but not with X intermediate. As product is being produced with Y and Z, but not with X, thus X is not getting converted to Y and further to the product. Hence, it must encode for enzyme converting X to Y.
- Mutant strain *c*⁻ having defective *c* gene is developing color with all X, Y and Z intermediate. As product is being produced in all the cases, it must not be involved in conversion of X to Y or Y to Z or Z to P. Hence, it must encode for enzyme converting substrate to X.
- Mutant strain *d*⁻ having defective *d* gene is developing color only with Z intermediate, and not with X and Y intermediate. As, product is being produced with Z only, but not with X or Y, that means either X is not getting converted to Y or Y is not getting converted to Z and further to the product. We have already found that enzyme converting X to Y is encoded by gene *b*, thus gene *d* must be coding for enzyme responsible for the conversion of Y to Z.

Answer (D)

74. Topic: Molecular Basis of Genetic Diseases

A substance or a factor like a microorganism, a form of radiation or chemical substance that is capable of producing an effect, the presence or absence of which results in disease or in more advanced form of disease is known as therapeutic agent.

- Antisense strand is the sequence of nucleotides and is the complement of message sense. When mRNA forms a duplex with a complementary antisense RNA sequence, translation is blocked.
- Almost all the enzymes are made up of proteins except ribozymes (RNA enzymes). Naturally occurring ribozymes catalyze only phosphodiester bond transfer, ribozymes synthesized *in vitro* have been designed to perform even wider biochemical reactions such as the formation of amide bonds.
- Aptamers are peptide molecules or oligonucleotide that binds to a specific target molecule. They are usually created by selecting them from a large random sequence pool, but natural aptamers also exist in riboswitches.
- Avidin is a tetrameric biotin-binding protein produced in the oviducts of birds, reptiles and amphibians and deposited in the whites of their eggs. It is believed to function as an antibiotic in the eggs of birds, reptiles and amphibians.

Answer (D)

75. Topic: RNA Interference

- Micro RNA (miRNA) helps in sequence-specific inactivation of mRNA and blocks translation of selected mRNA. miRNAs are micro RNAs and are small RNAs that play role in inhibiting translation of complementary strand.
- Small interfering RNA (siRNA), sequence-specific inactivation of mRNA. siRNAs are small, double stranded fragments that are formed when double stranded RNA initiates the response during RNA silencing.
- shRNA, a short hairpin RNA or small hairpin RNA is an artificial RNA molecule that can be used to silence target gene expression via RNA interference.
- Small nuclear RNA (snRNA) helps in RNA splicing.

While miRNA, siRNA and shRNA involved in silencing (sequence specific inhibition of gene expression), snRNA play a role in splicing RNA.

Answer (D)

76. Topic: Translation

The process by which amino acids are polymerized to form polypeptide is known as translation. In both prokaryotes and eukaryotes, the process of elongation is basically the same and elongation factors are required at various steps in the process. Polypeptide chain synthesis is divided into three main parts: a) initiation, b) elongation and c) termination.

Elongation of polypeptide chain has the following basic steps which are repeated as amino acids are polymerized into the growing polypeptide chain:

- Step 1: Amino acyl tRNA selection. Binding of an aminoacyl-tRNA to the A site of the ribosome. The tRNA binds to the mRNA by forming complementary base pairs with the tRNA anticodon.
- Step 2: Peptide bond formation. Transfer of the growing polypeptide chain from the P site to the A site by the formation of a new peptide bond. The elongating chain from the tRNA in the P site is uncoupled and covalently joins the chain to the tRNA in the A site. This reaction is catalyzed by the enzyme peptidyl transferase which is present in the 23S rRNA of 50 S subunit. Formation of a peptide bond between the amino group of the aminoacyl-tRNA in the A site and the carboxyl terminus of the growing polypeptide chain attached to the tRNA in the P site. Peptide bond formation requires the hydrolysis of the molecule of GTP.

- Step 3: Translocation of the ribosome along the mRNA from codon to codon. This helps to position the next codon in the A site.
- Step 4: Releasing deacylated tRNA.

Answer (C)

77. **Topic: Mutations and Mutagenesis**

Frame-shift mutation affects more than one base in DNA by a deletion or an insertion of one or more bases. It can alter all the three-base sequences beyond the deletion or insertion and is also called deletion or insertion mutation. In frameshift mutation, there is a change in reading frame by insertion or deletion of number of nucleotides that are not divisible by three. Since the gene repression by codons has a triplet nature, the translated product formed is completely different from original. The frameshift mutation also causes alteration in the first stop codon in the sequence.

- In (Q), g is added leading to change in reading frame.
- In (R), c is deleted leading to a change in reading frame.

Answer (B)

78. **Topic: Molecular Basis of Genetic Disease**

- In human beings, Huntington disease is an example for autosomal dominant inheritance that cause lethal expression. Individuals with the gene in heterozygous genotype survive upto 40 years but undergo gradual degeneration in brain and neural tissues.
- Traits that exist on the Y chromosome are Y-linked because they only occur on that chromosome and do not change in recombination and are known as Y-linked inheritance. For example: hairy ears.
- Cystic fibrosis is an autosomal recessive disorder, meaning is inherited when an individual receives a mutated copy of the gene associated with cystic fibrosis from both parents and is not inherited solely from the mother or father as in sex-linked disorders.
- Hemophilia is inherited in an *X*-linked recessive pattern. The genes associated with these conditions are located on the X chromosome, which is one of the two sex chromosomes. People having hemophilia (an X-linked blood-clotting disorder) are unable to produce a factor needed for blood clotting. It is caused due to a recessive X-linked mutation and it usually affects the males who have inherited the mutation from their heterozygous mothers. Affected males pass on the mutation to their daughter who usually does not develop hemophilia as they have inherited wild type allele from their mothers. However, affected males never transmit the mutant allele to their sons.

Answer (C)

79. **Topic: Mendelian Inheritance**

Crossing to be done between $DdEeFfgg \times ddEeFfGg$

As allelic pairs of all genes assort independently, thus,

(i) $Dd \times dd \rightarrow Dd, dd$ (Proportion of dd is 0.5)

(ii) $Ee \times Ee \rightarrow EE, Ee, eE, ee$ (Proportion of ee is 0.25)

(iii) $Ff \times Ff \rightarrow FF, Ff, fF, ff$ (Proportion of ff is 0.25)

(iv) $gg \times Gg \rightarrow Gg, gg$ (Proportion of gg is 0.5)

Thus, proportion of genotype $ddeeffgg$ from (i), (ii), (iii), (iv) would be:

$$= 0.5 \times 0.25 \times 0.25 \times 0.5 = 0.0156$$

Percentage of the proportion of progeny having the genotype $ddeeffgg = 0.0156 \times 100 = 1.56\%$.

Answer (1.56%)

80. **Topic: Molecular Structure of Genes and Chromosomes**

It is clear that it cannot be DNA as it has uracil and not thymine. So, it must be RNA and not DNA because DNA contains thymine instead of uracil. The base composition is not uniform, number of adenine nucleotide is not equal to the number of uracil and number of guanine nucleotide is not equal to that of cytosine. Hence it is single-stranded because in dsRNA, adenine would have been equal to uracil and cytosine would have been equal to guanine.

Answer (B)

81. **Topic: Translation**

Transfer RNA (tRNA) is a set of small, compact molecules that function to deliver amino acids to the ribosomes for assembly into proteins. They function as adaptors between amino acids and the codons in mRNA during translation (or protein synthesis). tRNA transports amino acids, which are used to build the protein, to the ribosomes; and position each amino acid in correct sequence on the elongating protein chain.

tRNAs are involved in translation of a sequence of mRNA codons into a sequence of amino acid residues. The ATP activates the amino acid by formation of an adenylated amino acid bound to aminoacyl-tRNA synthetase. Eventually transfer of amino acid to tRNA molecules takes place. The synthetase is thus responsible for 'charging' all the tRNAs that are suitable for that amino acid.

Hence, t-RNAs are involved in charging, initiation and elongation but not in termination because no tRNAs exist whose anticodons are complementary to a stop codon. Upon reaching stop codon (UAA, UAG or UGA), the ribosome reads the signal to stop further elongation and releases the polypeptide associated with the last tRNA.

Answer (D)

82. **Topic: Transcription**

The Dominant Base at each site is: AGTCACNTGC.

At the 3rd position distribution of guanine and cytosine is equal, thus we can take any of these, thus denoting it with N.

Answer (B)

83. Topic: Microbial Genetics

Size of genome = 6 million base pairs = 6×10^6 bp

Rate of DNA synthesis in sec = 1000 bp/sec

If, 1000 bp are synthesized in 1sec

6×10^6 bp are synthesized in $6 \times 10^6 / 10^3 = 6000$ sec

60 sec = 1 min

6000 sec = 6000 / 60 min = 100 min

Answer (100 min)

84. Topic: Transcription

Probability of getting any of the 4 nucleotides is equal and sums to to be one.

Thus, $p_A = p_T = p_G = p_C = \frac{1}{4} = 0.25$

Shanon Entropy

$$(S) = -\sum_{i=1}^{4} p_i \ln p_i$$

$$= -(p_A \ln p_A + p_T \ln p_T + p_G \ln p_G + p_C \ln p_C)$$

$$= -(0.25 \ln 0.25 + 0.25 \ln 0.25 + 0.25 \ln 0.25 + 0.25 \ln 0.25)$$

$$= -(4 \times -0.346) = 1.386$$

Answer (1.386)

85. Topic: Molecular Basis of Genetic Diseases

It is clear from the pedigree that the disease is getting transferred from father to only son. No female members of the family have been affected. Y-linked traits never occur in females and occur in all male descendants of an affected male. Thus, it is clearly a Y-linked disease. The concepts of dominant and recessive do not apply to Y-linked traits, as only one allele (on the Y) is present in all male individuals.

Answer (C)

86. Topic: Molecular Structure of Genes and Chromosomes

Circumference of a circle of radius, $r = 2\pi r$

Given, length of DNA = 25 nm

Then, $2\pi r = 25$

$$r = \frac{25}{2 \times 3.14} = 3.98 \text{ nm}$$

Answer (3.98 nm)

87. Topic: Molecular Structure of Genes and Chromosomes

The tips of each DNA molecule in a chromosome has an unusual stretch of repeated DNA sequences that along with a group of specialized proteins, form a "cap" at each end of a chromosome called a telomere. Human telomeres contain the sequence repeated from about 500 to 5000 times. The repeat sequence of telomere in humans is 5'-TTAGGG-3'. Most repeated sequence 3'-AATCCC-5' vary considerable but same telomere sequence is found in vertebrates. Most other organisms also have similar sequence.

Answer (B)

88. Topic: Transcription

A sense strand runs from 5' to 3' and is also called the coding strand of DNA. It is complementary to the template or antisense strand of DNA running from 3' to 5'. The sequence of mRNA formed during transcription is the same as sense strand since it uses antisense strand as the template. For a sense strand of DNA 5'-ATGGACCAGA-3', the resulting RNA sequence after transcription is 5'-AUGGACCAGA-3'.

The mRNA molecules are coding strands of RNA. They are also called sense strands of RNA because their nucleotide sequences make sense as they specify sequences of amino acids in protein when translation takes place.

Answer (D)

89. Topic: RNA Interference

- sno RNA are formed by splicing, debranching and trimming of introns of rRNA. They associate with proteins to form sno ribonucleoproteins (sno RNPs). They help in rRNA processing in the nucleolus. This processing is important for better stability and folding of rRNAs.
- piRNA, the largest class of small non-coding RNA molecules, protects germline from transposable elements (TEs) that can replicate and get inserted in genomic locations having both advantage in genome evolution as well as disadvantage in causing DNA breaks that might affect genome integrity. If there is excessive damage to the germ cell genome, it may even result in sterility. Therefore, piRNA pathway, an RNA silencing pathway controls TE activity in animal cells.
- miRNA are non-coding RNA molecules that bind to target mRNA by base pairing and prevent translation.
- snRNAs are small RNA molecules present in spliceosomes (RNA/protein complex) that carry out nuclear pre-mRNA splicing. There are five types of snRNA (U1, U2 U4, U5 and U6) that are involved in mRNA splicing as components of spliceosome.

Answer (D)

90. Topic: Molecular Structure of Genes and Chromosomes

Guanine and cytosine make up a complementary nitrogenous base pair as their available hydrogen bond donors and hydrogen bond acceptors pair with each other in space and makes three hydrogen bonds between bases. Similarly, adenine and thymine pair via hydrogen bond donors and

acceptors, but an A–T base pair has only two hydrogen bonds between the bases. Thus, association constant for G–C base pair must be higher than for A–T base pair.

Answer (B)

91. Topic: Molecular Structure of Genes and Chromosomes

DNA molecule has over all net negative charge because of the presence of phosphate groups. Out of the given amino acids, lysine (Lys), arginine (Arg) and glutamine (Gln) are basic amino acids and glutamate (Glu) is acidic amino acid. Arg has five hydrogen donors groups, thus attracts more phosphate and form hydrogen bonding with DNA and stabilizes the structure of DNA-protein. Therefore, they will be holding highest frequency in the graph. It will be followed by Lys holding second place because of its basic nature and number of hydrogen donors. Gln will hold third place as it has less hydrogen donors. And, Glu because of its negative charge will have least frequency in the graph.

Answer (A)

92. Topic: Molecular Basis of Genetic Diseases

Other than determining sex of the offspring, the sex chromosomes are also responsible for transmission of several non-sexual traits. Many genes are present on X chromosomes but are absent from Y chromosomes. This pattern of heredity is called sex-linked inheritance.

In the given question, female is the carrier but is not affected. Hence, it is a recessive trait. Had the trait been dominant, then even in the presence of a single allele, the disease could have affected the individual. If it is autosomal recessive, then genotype of affected male would be *dd*, one affected allele is coming from carrier mother, but the father is unaffected and thus there is no source of the other affected allele. Thus, the disease is not autosomal. It is hemizygous disease that is getting transferred from mother to son and is never passed from father to son. Therefore, the disease is X-linked recessive.

Answer (C)

93. Topic: Transcription

Eukaryotic mRNAs have a cap structure consisting of a 7-methylguanosine (m^7G) residue joined to the transcript's initial (5′) nucleotide via a 5′–5′ triphosphate bridge. The cap, which adds to the growing transcript when it is ~30 nt long, identifies the eukaryotic translation start site. Capping involves several enzymatic reactions:

- The removal of the leading phosphate group from the mRNA's 5′ terminal triphosphate group by an RNA triphosphatase.
- The guanylylation of the mRNA by capping enzyme, which requires GTP and yields the 5′-5′ triphosphate bridge and PP_i.
- The methylation of guanine by guanine-7-methyltransferase, in which the methyl group is supplied by S-adenosylmethionine (SAM).

Answer (C)

94. Topic: Mutations and Mutagenesis

- The variant polypeptide so produced will be unable to function in the cell. It will be released, and the appropriate protein will not be synthesized. The effect can be lethal if the missing protein is essential to cell structure or function. It is also called nonsense mutation.
- A synonymous mutation is a change in the DNA sequence that codes for amino acids in a protein sequence but does not change the encoded amino acid.
- On changing the sequence of the peptide, if the new protein functions poorly, the effect will be significant. Rarely, the new protein may function better and produce a phenotype that is better adapted to its environment than the original phenotype. This known as missense mutation.
- The mutation is said to be as same-sense mutation or silent mutation if amino acid replaced has no effect on the protein being synthesized.

Answer (B)

95. Topic: Microbial Genetics

Gene transfer refers to the movement of genetic information between organisms. It is significant because it greatly increases the genetic diversity of organisms. When genes pass from parents to offspring, this is called vertical gene transfer. Furthermore, bacteria can also do horizontal, or lateral, gene transfer, when they pass genes to other microbes of their same generation. There are three mechanisms of lateral gene transfer in bacteria—transformation, transduction, and conjugation.

- Transformation is a form of genetic recombination in which a DNA fragment from a dead, degraded bacterium enters a competent recipient bacterium and is exchanged for a piece of DNA of the recipient.
- Transduction involves the transfer of a DNA fragment from one bacterium to another by a bacteriophage. There are two forms of transduction: generalized transduction and specialized transduction.
- Conjugation is genetic recombination in which there is a transfer of DNA from a living donor bacterium to a living recipient bacterium by cell-to-cell contact.

While, transfection is the process of inserting genetic material, such as DNA and double stranded RNA, into mammalian cells.

Answer (C)

96. Topic: Mendelian Inheritance

Alpha 1-3-N-acetylgalactosaminyltransferase and alpha 1-3-galactosyltransferase genes encode proteins related to the first discovered blood group system, ABO. The allele present in an individual determines their blood group. The 'O' blood group is caused by a deletion of guanine-258 near the N-terminus of the protein which results in a frameshift and translation of an almost entirely different protein. Individuals with the A, B, and AB alleles express glycosyltransferase activities that convert the H antigen into the A or B antigen.

Answer (B)

97. Topic: Transcription

The starting point for capping with 7-methylguanylate is the unaltered 5′ end of an mRNA molecule, which terminates at a triphosphate group. This features a final nucleotide followed by three phosphate groups attached to the 5′ carbon. The capping process is initiated before the completion of transcription, as the nascent pre-mRNA is being synthesized. has an unusual 5′-5′ triphosphate linkage and two or more methyl groups.

Answer (C)

98. Topic: Mendelian Inheritance

It is given that the genetic cross between two purple flower-bearing plants is giving white flowers, it means that the purple flowers are having heterozygous genotype (*Pp*).

Thus, on crossing of two purple flowers with heterozygous genotype (*Pp*) would give,

	P	*p*
P	*PP* Purple	*Pp* Purple
p	*Pp* Purple	*Pp* White

Purple flowers (0.75): White (0.25)

Thus, probability of next offspring to have purple flowers is 0.75.

Answer (0.75)

CHAPTER 6 Analytical Techniques

SYLLABUS

Principles of Microscopy— Light, Electron, Fluorescent and Confocal; Centrifugation— High Speed and Ultra; Principles of Spectroscopy— UV, Visible, CD, IR, FTIR, Raman, MS, NMR; Principles of Chromatography— Ion Exchange, Gel Filtration, Hydrophobic Interaction, Affinity, GC, HPLC, FPLC; Electrophoresis; Microarray.

CHAPTER ANALYSIS

Topic	GATE 2010	GATE 2011	GATE 2012	GATE 2013	GATE 2014	GATE 2015	GATE 2016	GATE 2017	GATE 2018	GATE 2019
Principles of Microscopy: Light, Electron, Fluorescent and Confocal			1							
Principles of Centrifugation: High Speed and Ultra							1			
Principles of Spectroscopy: UV, Visible, CD, IR, FTIR, Raman, MS, NMR	1	2	1				3	3	2	2
Principles of Chromatography: Ion Exchange, Gel Filtration, Hydrophobic Interaction, Affinity, GC, HPLC, FPLC		1		1	1		1	2	1	
Electrophoresis										1
Microarray	1									

QUESTIONS

1. Gel retardation assay is routinely used to monitor interactions between
   (A) proteins.
   (B) drug and nucleic acid.
   (C) nucleic acids.
   (D) protein and nucleic acid.
   (GATE 2001; 1 Mark)

2. Cibacron Blue dye affinity chromatography can be used for affinity purification of
   (A) NADPH dehydrogenase.
   (B) glucoamylase.
   (C) subtilisin.
   (D) caspase.
   (GATE 2005; 1 Mark)

3. A protein binds to phosphocellulose column at pH 7.0 and elutes at pH 8.0. If the protein has to be further purified on a DEAE Sephacel column, the binding buffer should have a pH of
   (A) 5 (B) 6
   (C) 7 (D) 8
   (GATE 2007; 2 Marks)

4. Determine the correctness or otherwise of the following Assertion (*A*) and the Reason (*R*).
   Assertion: MTT assay is used to determine cell viability based on the principle of color formation by DNA fragmentation.
   Reason: MTT assay is used to determine cell viability based on the color development by converting tetrazolium soluble sail to insoluble salt.

(A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(B) Both (*A*) and (*R*) are true and (*R*) is not the correct reason for (*A*).
(C) (*A*) is true but (*R*) is false.
(D) (*A*) is false but (*R*) is true.

(GATE 2010; 2 Marks)

5. Match Group I with Group II.

Group I	Group II
P. Real Time-PCR	1. Biochips
Q. 2-D Electrophoresis	2. Syber Green
R. Affinity chromatography	3. Antibody linked sephrose beads
S. Microarray	4. Ampholytes

(A) P–1; Q–2; R–4; S–3
(B) P–2; Q–3; R–4; S–1
(C) P–2; Q–4; R–3; S–1
(D) P–3; Q–2; R–1; S–4

(GATE 2010; 2 Marks)

6. Hydrated synthetic seeds which are produced by ion exchange reaction involve mixing the somatic embryos in a solution of
(A) sodium alginate and dropping it in a solution of calcium nitrate.
(B) calcium alginate and dropping it in a solution of sodium nitrate.
(C) calcium alginate and dropping it in a solution of ammonium nitrate.
(D) mannitol and dropping it in a solution of sodium nitrate.

(GATE 2011; 1 Mark)

7. Match the items in Group I with Group II.

Group I	Group II
P. Circular dichroism	1. Concentration
Q. X-ray crystallography	2. Sedimentation coefficient
R. Freeze-drying	3. Secondary structure determination
S. Ultracentrifugation	4. Tertiary structure determination

(A) P–4; Q–1; R–2; S–3
(B) P–1; Q–4; R–3; S–2
(C) P–2; Q–3; R–4; S–1
(D) P–3; Q–4; R–1; S–2

(GATE 2011; 2 Marks)

8. Protein-protein interactions are studied by
P. DNA foot printing
Q. Yeast two hybrid system
R. Ligase chain reaction
S. Mass spectrometry
(A) P and S only
(B) Q and S only
(C) P and R only
(D) Q and R only

(GATE 2011; 2 Marks)

9. In transmission electron microscopy, electron opacity is greatly enhanced by treating the specimen with
(A) ferrous ammonium sulfate.
(B) uranium acetate.
(C) sodium chloride.
(D) basic fuchsin.

(GATE 2012; 1 Mark)

10. The helix content of a protein can be determined using
(A) an infrared spectrometer.
(B) a fluorescence spectrometer.
(C) a circular dichroism spectrometer.
(D) a UV-Visible spectrophotometer.

(GATE 2012; 1 Mark)

11. Match the entries in the Group I with the elution conditions in Group II.

Group I	Group II
P. Ion exchange chromatography	1. Isocratic solvent
Q. Hydrophobic column chromatography	2. Ampholytes
R. Gel filtration chromatography	3. Increasing gradient of salt
S. Chromatofocusing	4. Decreasing gradient of polarity

(A) P–4; Q–1; R–2; S–3
(B) P–4; Q–3; R–1; S–2
(C) P–3; Q–4; R–1; S–2
(D) P–3; Q–4; R–2; S–1

(GATE 2013; 2 Marks)

12. If the dissociation constant for solute-adsorbent binding is K_D, the retention time of the solute in a chromatography column
(A) increases with increasing K_D
(B) decreases with increasing K_D
(C) passes through minimum with increasing K_D
(D) is independent of K_D

(GATE 2014; 1 Mark)

13. Cesium chloride density gradient centrifugation is commonly used for the separation of DNA molecules. The buoyant density, ρ, of a double stranded Cs$^+$ DNA is given by the equation $\rho = 1.66 + 0.098\ X_{G+C}$, where X_{G+C} denotes

(A) total number of G and C.
(B) mole fraction of G + C.
(C) number of GC repeats.
(D) ratio of G + C to A + T content

(GATE 2016; 1 Mark)

14. Which one of the following techniques relies on the spin angular momentum of a photon?

(A) CD spectroscopy
(B) Fluorescence spectroscopy
(C) IR spectroscopy
(D) Raman spectroscopy

(GATE 2016; 1 Mark)

15. The molar extinction coefficients of Trp and Tyr at 280 nm are 5690 and 1280 $M^{-1}cm^{-1}$, respectively. The polypeptide chain of yeast alcohol dehydrogenase (37kDa) contains 5 Trp and 14 Tyr residues. The absorbance at 280 nm of a 0.32 mg mL^{-1} solution of yeast alcohol dehydrogenase measured in a cuvette of 1 cm pathlength will be _____.

(Assume that the molar extinction coefficient values for Trp and Tyr apply to these amino acids in the yeast alcohol dehydrogenase).

(GATE 2016; 2 Marks)

16. The activity of lactate dehydrogenase can be measured by monitoring the following reaction:

$$\text{Pyruvate} + \text{NADH} \rightarrow \text{Lactate} + \text{NAD}^+$$

The molar extinction coefficient of NADH at 340 nm is 6220 $M^{-1}cm^{-1}$. NAD^+ does not absorb at this wavelength. In an assay, 25 µL of a sample of enzyme (containing 5 µg protein per mL) was added to a mixture of pyruvate and NADH to give a total volume of 3 mL in a cuvette of 1 cm pathlength. The rate of decrease in absorbance at 340 nm was 0.14 min^{-1}. The specific activity of the enzyme will be _____p mol $min^{-1}mg^{-1}$.

(GATE 2016; 2 Marks)

17. A protein is to be purified using ion-exchange column chromatography. The relationship between HETP (Height Equivalent to Theoretical Plate) and the linear liquid velocity of mobile phase is given by:

$$H = \frac{A}{u} + Bu + C$$

where, H is HETP (m) and u is linear liquid velocity of mobile phase (m s^{-1}). The values of A, B and C are 3×10^{-8} m^2 s^{-1}, 3s and 6×10^{-5} m, respectively. The number of theoretical plates based on minimum HETP for a column of 66 cm length will be _____.

(GATE 2016; 2 Marks)

18. In a thin layer chromatography experiment using a silica gel plate, a compound showed migration of 12.5 cm and the solvent front showed migration of 18 cm. The R_f value for the compound is _____.

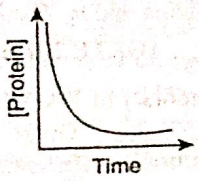
(GATE 2017; 1 Mark)

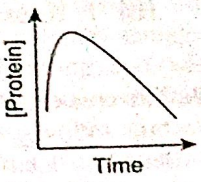
19. Which one of the following techniques can be used to determine the structure of a 15 kDa globular protein at atomic resolution?

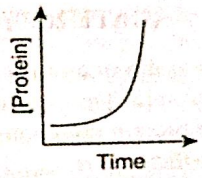
(A) Raman spectroscopy
(B) IR spectroscopy
(C) UV spectroscopy
(D) NMR spectroscopy

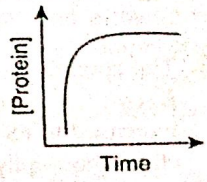
(GATE 2017; 1 Mark)

20. Which one of the following graphs represents the kinetics of protein precipitation by addition of ammonium sulphate? On the Y-axis, [Protein] represents the concentration of free protein in solution.

(A)


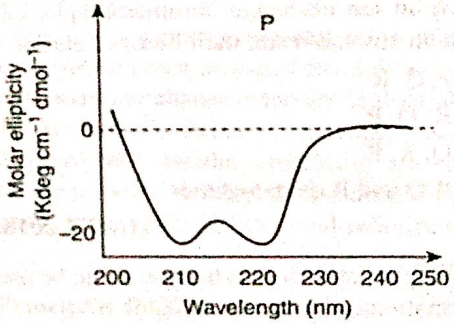
(B)


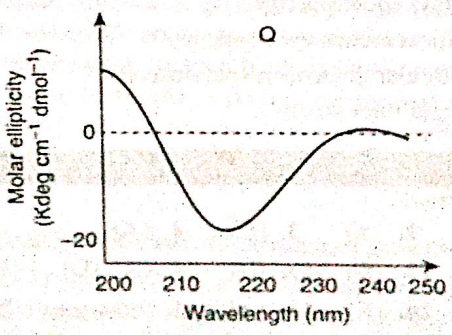
(C)


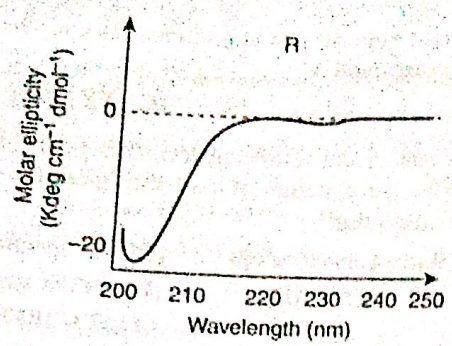
(D)


(GATE 2017; 2 Marks)

21. The circular dichroism spectra of three proteins P, Q, and R are given below:







Choose the correct combination.

(A) P: α-helix, Q: β-sheet and R: Random coil
(B) P: β-sheet, Q: α-helix and R: Random coil
(C) P: α-helix, Q: Random coil and R: β-sheet
(D) P: Random coil, Q: α-helix and R: β-sheet

(GATE 2017; 2 Marks)

22. The concentration (in micromolar) of NADH in a solution with $A_{340} = 0.50$ is _____. Given data: Path length = 1 cm: Molar extinction coefficient of NADH $\varepsilon_{340} = 6220\ M^{-1}cm^{-1}$

(GATE 2017; 2 Marks)

23. The specific activity of an enzyme in a crude extract of *E. coli* is 9.5 units mg^{-1} of protein. The specific activity increased to 68 units mg^{-1} of protein upon ion-exchange chromatography. The fold purification is _____.

(GATE 2017; 2 Marks)

24. A mixture contains three similarly sized peptides P, Q and R. The peptide P is positively charged, Q is weakly negative, and R is strongly negative. If this mixture is passed through an ion-exchange chromatography column containing an anionic resin, their order of elution will be

(A) P, Q, R
(B) R, Q, P
(C) Q, R, P
(D) P, Q and R elute together

(GATE 2018; 1 Mark)

25. Which of the following technique(s) can be used to study conformational changes in myoglobin?

P. Mass spectrometry
Q. Fluorescence spectroscopy
R. Circular dichroism spectroscopy
S. Light microscopy

(A) P only **(B)** P and S only
(C) Q and R only **(D)** S only

(GATE 2018; 1 Mark)

26. The absorbance of a solution of tryptophan measured at 280 nm in a cuvette of 2.0 cm path length is 0.56 at pH 7. The molar extinction coefficient (ε) for tryptophan at 280 nm is 5600 $M^{-1}cm^{-1}$ at pH 7. The concentration of tryptophan (in μM) in the solution is _____.

(GATE 2018; 1 Mark)

27. A UV-visible spectrophotometer has a minimum detectable absorbance of 0.02. The minimum concentration of a protein sample that can be measured reliably in this instrument with a cuvette of 1 cm path length is _____ μM. The molar extinction coefficient of the protein is 10,000 $L\ mol^{-1}\ cm^{-1}$.

(GATE 2019; 2 Marks)

28. Group I lists spectroscopic methods and Group II lists bimolecular applications of these methods. Match the methods in Group I with the applications in Group II.

Group I	Group II
P. Infrared	1. Identification of functional groups
Q. Circular Dichroism	2. Determination of secondary structure
R. Nuclear Magnetic Resonance	3. Estimation of molecular weight
	4. Determination of 3-D structure

(A) P–4; Q–3; R–1 **(B)** P–2; Q–1; R–3
(C) P–1; Q–2; R–4 **(D)** P–3; Q–2; R–4

(GATE 2019; 2 Marks)

29. The hexapeptide P has an isoelectric point (pI) of 6.9. Hexapeptide Q is a variant of P that contains valine instead of glutamate at position 3. The two peptides are analyzed by polyacrylamide gel electrophoresis at pH 8.0. Which one of the following statements is CORRECT?

(A) P will migrate faster than Q towards the anode.
(B) P will migrate faster than Q towards the cathode.
(C) Both P and Q will migrate together.
(D) Q will migrate faster than P towards the anode.

(GATE 2019; 2 Marks)

ANSWER KEY

1. (D) **2.** (A) **3.** (C) **4.** (A) **5.** (C) **6.** (A) **7.** (D) **8.** (B) **9.** (B) **10.** (C)
11. (B) **12.** (B) **13.** (B) **14.** (A), (D) **15.** (0.37) **16.** (540 $\mu mol\ min^{-1}mg^{-1}$) **17.** (1000) **18.** (0.69)
19. (D) **20.** (A) **21.** (A) **22.** (80.4 μm) **23.** (7.16) **24.** (B) **25.** (C) **26.** (50 μM) **27.** (2 μM) **28.** (C)
29. (A)

ANSWERS WITH EXPLANATIONS

1. **Topic: Electrophoresis**

The gel retardation assay, also known as the band shift assay or electrophoretic mobility shift assay (EMSA), is used to detect DNA-binding proteins in crude cell extracts and to study the binding activity of purified DNA-binding proteins. It is also applicable to the study of RNA-protein interactions. The basis of the assay is that protein-DNA complexes remain intact when gently fractionated by gel electrophoresis and migrate as distinct bands more slowly than the free DNA fragment.

Answer (D)

2. **Topic: Principles of Chromatography**

The dye Cibacron Blue F3GA has a high affinity for many proteins and enzymes. It has therefore been attached to various solid supports such as sephadex, sepharose, polyacrylamide, and the like. In the immobilized form the dye has rapidly been exploited as an affinity chromatographic medium to separate and purify a variety of proteins including dehydrogenases, kinases, serum albumin, interferons, several plasma proteins, and a host of other proteins.

Answer (A)

3. **Topic: Principles of Chromatography**

DEAE Sephacel is a weak anion exchanger based on beaded cellulose. The ion exchange group is diethylaminoethyl, which remains charged and maintains consistently high capacity over the entire working range, pH 2–9. It is macroporous and has an exclusion limit of approximately 1,000,000 daltons for globular proteins. During purification of protein on a DEAE Sephacel column, the binding buffer should have a pH of 7, same as for phosphocellulose column.

Answer (C)

Topic: Principles of Spectroscopy

An MTT assay is a colorimetric assay based on assessing the cell metabolic activity. The biochemical mechanism behind the MTT assay involves NAD(P)H-dependent cellular oxidoreductase enzyme that converts the yellow tetrazolium MTT [3-(4, 5-dimethylthiazolyl-2)-2,5-diphenyltetrazolium bromide] into insoluble (*E,Z*)-5-(4,5-dimethylthiazol-2-yl)-1,3-diphenylformazan (formazan). The formed formazan can be dissolved with dimethyl sulfoxide (DMSO) to give a purple color with characteristic absorption at 540 nm. Intensity of purple color is directly proportional to the cell number and thus indicating the cell viability.

Answer (A)

5. **Topic: Principles of Chromatography, Electrophoresis and Microarray**

- Real Time PCR is based on the detection of the fluorescence produced by a reporter molecule, which increases as the reaction proceeds. This occurs due to the accumulation of the PCR product with each cycle of amplification. These fluorescent reporter molecules include dyes that bind to the double-stranded DNA such as SYBR green or sequence specific probes such as TaqMan Probes.
- 2D gel electrophoresis is a key technique for purifying individual proteins from complex samples based on their charge (isoelectric focusing) and their molecular weights (SDS-PAGE). This technique is based on the use of carrier ampholytes that are small molecules with different carboxylic and amino groups. These are mixed with a polymerization mix, poured into a glass capillary tube and allowed to solidify. During the pre-run, the carrier ampholyte units will migrate according to their charge and reach their respective pI value and create a pH gradient that can be used for isoelectric focusing.
- Affinity chromatography separates proteins on the basis of a reversible interaction between a protein (and group of proteins) and a specific ligand coupled to a chromatographic matrix. Sepharose, a bead-form of agarose, provides many of the properties required in an ideal matrix.
- The biochip is a microarray that is, a collection of miniaturized test sites, arranged on a solid substrate that permits many simultaneous tests of biological material to be performed, through screening miniaturized, multiplexed and parallel processing and detection. They have been developed to enable large-scale genomic, proteomic and functional genomic analyses.

Answer (C)

6. **Topic: Principles of Chromatography**

2–3% sodium alginate is poured drop-wise (containing inert embryo) in calcium nitrate solution for encapsulation of the synthetic seed. Sodium is immediately replaced by calcium which forms the cross-linking between alginate polymers and thus produced meshwork structure and forms a gel layer around the artificial seed.

Answer (A)

7. **Topic: Principle of Centrifugation and Spectroscopy**

- The CD spectrum of a protein in the "near-UV" spectral region (250–350 nm) can be sensitive to certain aspects of secondary structure.

- X-Ray Crystallography is a technique for examining the structure of crystal and describe an atomic structure.
- The sedimentation coefficient 'S' of a particle is used to describe its behavior in sedimentation processes, usually ultracentrifugation.
- Freeze drying (lyophilization or cryodesiccation) involves a low temperature dehydration process which freezes the product by lowering pressure and then removing the ice by sublimation.

Answer (D)

9. Topic: Principles of Spectroscopy

- DNA foot printing is a technique of examining the sequence specificity of DNA-binding proteins in vitro.
- The yeast 2-hybrid (Y2H) technique is well known to detect protein-protein interactions.
- Ligase chain reaction allows quick amplification of DNA.
- Mass spectrometry is a powerful analytical technique that ionizes chemical species and sorts the ions based on their mass-to-charge ratio. It is currently at the head of technologies for mapping protein-protein interactions, as it is a highly sensitive technique.

Answer (B)

10. Topic: Principles of Microscopy

Heavy metals such as uranium and lead are the most widely used stains for electron microscopy. Uranium acetate produces the highest contrast, electron density and imparts a fine grain to the image. Uranyl ions attach to DNA or RNA through their phosphate groups and to proteins and lipids by oxalic acid carboxyl groups.

Answer (B)

10. Topic: Principles of Spectroscopy

The helix content of a protein can be determined using a circular dichroism spectrometer. It is based on the absorption of right-hand circular or left-hand circular polarized light by an optically active molecule. Hence, it is shown by dextrorotary and levorotary components of the biomolecule. In addition, it is also able to differentiate between secondary structures of proteins such as alpha helix as well as double helices of DNA or RNA.

Answer (C)

11. Topic: Principles of Chromatography

- Ion chromatography is a type of column chromatography, that divides ions and polar molecules based on their affinity to the ion exchanger.
- Hydrophobic interaction chromatography (HIC) separates proteins based on the hydrophobicity of the protein. Hydrophobic interactions tend to be strongest at high salt concentrations. After binding at high salt, the protein is eluted using a decreasing salt gradient.
- Gel filtration chromatography (size exclusion chromatography) separates proteins, peptides or oligonucleotides on the basis of size and in some through their weight. Isocratic elution (same buffer throughout) is used in this.
- Chromatofocusing combines the advantages of high-capacity ion-exchange procedures with the high resolution of isoelectric focusing into a single chromatographic focusing procedure. It is a powerful separation method that allows resolutions of single proteins and other ampholytes (molecules that contain both acidic and basic group) from a complex mixture according to differences in their isoelectric point.

Answer (B)

12. Topic: Principles of Chromatography

Retention time (RT) is a measure of the time taken for a solute to pass through a chromatography column. A solute's retention time or retention volume in chromatography is directly related to the strength of the solute's interaction with the mobile and stationary phases. Thus, more is the dissociation constant (K_d), less strength in the interaction and less would be the retention time.

Answer (B)

13. Topic: Principles of Centrifugation

X represents mole fraction of G + C. CsCl is better suited for routine determination of the guanine + cytosine (G + C) content of DNA, because in this solvent, there is an almost linear relationship between the percentage of G + C (20–80 %) and the buoyant density of DNA.

Answer (B)

14. Topic: Principles of Spectroscopy

- CD is observed when optically active matter absorbs left- and right-circular polarized light slightly differently. The CD spectra provide valuable information about secondary structure of biological macromolecules, rapid structural comparisons and molecular interactions. CD is often referred to as dichroism for spin angular momentum as left-hand and right-hand circular polarized light represents two possible states of spin angular momentum of proton.
- Fluorescence is the phenomenon in which absorption of light of a given wavelength by a molecule (excitation) is followed by the emission of light at longer wavelengths.

- The absorption spectroscopy depends upon the type of transition involved and accordingly the frequency range of the electromagnetic radiation absorbed. If the transition is from one vibrational energy level to another then the radiation is from the infrared portion of the electromagnetic spectrum and the technique is known as infrared (IR) spectroscopy.
- Raman spectroscopy is based on inelastic scattering of monochromatic light which can be from laser visible, near IR (infra-red) or near UV (ultra-violet) range. The energy of laser photons shifts up or down which provides information about vibrational modes in the system.

Answer (A), (D)

15. Topic: Principles of Spectroscopy

Given,

Wavelength, $\lambda = 280$ nm

Molecular mass of alcohol dehydrogenase

$= 37\text{kDa} = 37 \times 10^3\ \text{Da} = 37 \times 10^3\ \text{g}$

Number of Trp residue in alcohol dehydrogenase = 5

Number of Tyr residue in alcohol dehydrogenase = 14

Molar extinction coefficient of Trp, $\varepsilon_{Trp} = 5690\ \text{M}^{-1}\text{cm}^{-1}$

Molar extinction coefficient of Tyr, $\varepsilon_{Tyr} = 1280\ \text{M}^{-1}\text{cm}^{-1}$

Total molar extinction coefficients $= 5\varepsilon_{Trp} + 14\varepsilon_{Tyr} = \varepsilon_{max}$

$\varepsilon_{max} = 5(5690) + 1280 = 46370\ \text{M}^{-1}\text{cm}^{-1}$

Solution concentration $= 0.32\ \text{mg mL}^{-1} = 0.32\ \text{g L}^{-1}$

To find concentration of enzyme (c),

$37 \times 10^3\ \text{g L}^{-1}$ of enzyme $= 1$ mol

$0.32\ \text{g L}^{-1}$ of enzyme $= \dfrac{0.32}{37 \times 10^3} = 0.86 \times 10^{-5}$ mol

Length of light path, $l = 1$ cm

We know,

$$A = \varepsilon c l$$
$$= 46370 \times 0.8 \times 10^{-5} \times 1$$
$$= 0.37$$

Answer (0.37)

16. Topic: Principles of Spectroscopy

Given,

Wavelength, $\lambda = 340$ nm

Molar extinction coefficient of NADH,

$\varepsilon_{NADH} = 6220\ \text{M}^{-1}\text{cm}^{-1}$

Molar extinction coefficient of NAD$^+$, $\varepsilon_{NAD^+} = 0$

Length of light path, $l = 1$ cm

Total volume, $V_{total} = 3$ mL

Volume of sample, $V_{sample} = 25\mu\text{L} = 25 \times 10^{-3}$ mL

Now, 1 mL contains $= 5\ \mu\text{g} = 5 \times 10^{-3}$ mg protein

So, 25×10^{-3} mL contains $= 5 \times 10^{-3} \times 25 \times 10^{-3}$

$= 125 \times 10^{-6}$ mg protein

Concentration of protein, $[E] = 125 \times 10^{-6}$ mg (i)

Rate of decrease in absorbance, $\dfrac{dA}{dt} = 0.14\ \text{min}^{-1}$

We know,

$$A = \varepsilon c l$$
$$\frac{dA}{dt} = \varepsilon l \frac{dc}{dt}$$
$$\frac{dc}{dt} = \frac{dA}{dt} \times \frac{1}{\varepsilon l} = \frac{0.14}{6220 \times 1}\ \text{M min}^{-1}$$
$$\frac{dc}{dt} = \frac{0.14}{6220 \times 1}\ \text{mol L}^{-1}\ \text{min}^{-1}$$
$$= \frac{0.14 \times 10^6}{6220 \times 1 \times 10^3}\ \mu\text{mol mL}^{-1}\ \text{min}^{-1}$$
$$\frac{dc}{dt} = \frac{0.14 \times 10^3}{6220}\ \mu\text{mol mL}^{-1}\ \text{min}^{-1} \quad \text{(ii)}$$

We know, specific activity of enzyme, $\sigma_{s.a.} = \dfrac{dc}{dt} \times \dfrac{V_{total}}{[E]}$

From (i) and (ii)

$$\sigma_{s.a.} = \frac{0.14 \times 10^3}{6220}\ \mu\text{mol mL}^{-1}\ \text{min}^{-1} \times \frac{3\text{mL}}{125 \times 10^{-6}\ \text{mg}}$$

$\sigma_{s.a.} = 540\ \mu\text{mol min}^{-1}\ \text{mg}^{-1}$

Answer (540 μmol min^{-1}mg^{-1})

17. Topic: Principles of Chromatography

Given,

$$H = \frac{A}{u} + Bu + C$$

H = Plate height

u = Linear liquid velocity

$A = 3 \times 10^{-4}\ \text{m}^2\ \text{s}^{-1}$

$B = 3s$

$C = 6 \times 10^{-3}$ m

For minimum HETP,

$$u = \sqrt{\frac{A}{B}} = \sqrt{\frac{3 \times 10^{-4}\ \text{m}^2\text{s}^{-1}}{3s}} = 10^{-4}\ \text{ms}^{-1}$$

Therefore,

$$H = \frac{3 \times 10^{-4}\ \text{m}^2\text{s}^{-1}}{10^{-4}\ \text{ms}^{-1}} + 3s \times 10^{-4}\ \text{ms}^{-1} + 6 \times 10^{-3}\ \text{m}$$
$$= 3 \times 10^{-4}\ \text{m} + 3 \times 10^{-4}\ \text{m} + 6 \times 10^{-3}\ \text{m}$$
$$= 6.6 \times 10^{-3}\ \text{m}$$

Given, Length of column, $L = 66$ cm $= 66 \times 10^{-2}$ m

Number of theoretical plates can be calculated using formula

$$N = \frac{L}{H} = \frac{8.8 \times 10^{-3}\ m}{8.8 \times 10^{-6}\ m} = 1000\ plates$$

Answer (1000)

18. Topic: Principles of Chromatography

R_f is the "Retardation Factor", which is the ratio of the distance travelled by a compound in a mobile phase to the distance travelled by the front of mobile phase.

$$R_f = \frac{\text{Migration distance of substance}}{\text{Migration distance of solvent front}}$$

$$R_f = \frac{12.5}{18} = 0.69$$

Answer (0.69)

19. Topic: Principles of Spectroscopy

- Nuclear magnetic resonance (NMR) provides useful structural information about the position of linkages, anomeric carbon configuration and about the sequence of oligosaccharides. NMR is used to determine the atomic resolution structures of proteins. This is done by obtaining a separate signal for each atom in the protein. Globular proteins reorient rapidly in solution and their structure can be established by NMR spectroscopy.
- Raman spectroscopy is used to identify the molecules.
- IR spectroscopy is mainly used to determine functional groups in a molecule.
- UV spectroscopy is commonly used to determine conjugated organic compounds, transition metal ions, etc., quantitatively.

Answer (D)

20. Topic: Analytic Techniques

When ammonium sulphate salt is added to the solution, the surface tension of the water increases, resulting in increased hydrophobic interaction between protein and water. The protein responds to this situation by decreasing its surface area in an attempt to minimize contact with the solvent—as manifested by folding (the folded conformation is more compact than the unfolded one) and then self-association leading to precipitation. Thus, on addition of ammonium sulphate, amount of free protein would decrease with time. This is depicted in Graph (A).

Answer (A)

21. Topic: Principle of Spectroscopy

- α-helical proteins have negative bands at 222 nm and 208 nm and a positive band at 193 nm.
- Proteins with well-defined antiparallel β-pleated sheets (β-helices) have negative bands at 218 nm and positive bands at 195 nm.
- Unordered proteins have very low ellipticity above 210 nm and negative bands near 195 nm.

Thus, P is representing α-helix protein, Q is representing β-sheet and R is representing Random coil.

Answer (A)

22. Topic: Principles of Spectroscopy

Absorbance = 0.50

Path length = 1 cm

Extinction coefficient (ε) = 6220 M^{-1} cm^{-1}

By Beer's law: $A = \varepsilon cl$

$$c = \frac{A}{\varepsilon l} = \frac{0.5}{6220 \times 1} = 80.4\ \mu m$$

Answer (80.4 μm)

23. Topic: Principles of Chromatography

Initial specific activity = 9.5 units mg^{-1}

Final specific activity = 68 units mg^{-1}

$$\text{Fold purification} = \frac{\text{Final specific activity}}{\text{Initial specific activity}} = \frac{68}{9.5} = 7.16$$

Answer (7.16)

24. Topic: Principle of Chromatography

Ion-exchange chromatography works on the principle of electrostatic attraction between species of opposite charge. In ion-exchange chromatography, ionic bonding of proteins to an inert matrix material takes place. Ion-exchange resins are cross-linked polymers containing ionic group as part of their structure. Binding of proteins is reversible and can be displaced by increasing the buffers ionic strength or altering its pH. Elution of proteins from the column takes place in the order from that bound least strongly to that bound most strongly. When various peptides run through an anionic resin that is negatively charged, they will be eluted in order from strong negative charge to positive charge. So, the order of elution will be R (strongly negative), Q (weakly negative) and P (positive) peptides.

Answer (B)

25. Topic: Principles of Spectroscopy

- Electrospray ionization mass spectrometry (ESIMS) has made it possible to employ mass spectrometry for determination of molecular masses, chemical formula and molecular structure of proteins.
- Fluorescence microscopy is used to study protein structure and functions.
- Circular dichroism spectroscopy is used to determine protein folding and understanding its secondary and tertiary structures. Circular dichroism and fluorescence spectroscopy complemented with ESIMS, is used to study the mechanism of acid- and organic solvent-induced denaturation by conformational changes in Mb.

- Light microscopy is used to examine small objects using visible light and magnifying lenses.

Answer (C)

26. Topic: Principles of Spectroscopy

Absorbance (A) = $\varepsilon c l$

where, ε (molar extinction coefficient) for tryptophan at 280 nm = 5600 $M^{-1}cm^{-1}$

l (path length) = 2.0 cm

A (absorbance) = 0.56

Putting this in formula we get

$$0.56 = 5600 \times c \times 2$$

$$c = \frac{0.56}{5600 \times 2} = 5 \times 10^{-5} M$$

$$c = 50 \times 10^{-6} M = 50 \mu M$$

Answer (50 μM)

27. Topic: Principle of Spectroscopy

According to Beer-Lambert law,

$$A = \varepsilon c l$$

A, absorbance (unitless, $A = \log_{10} P_0 / P$) = 0.02

ε, molar absorptivity (L mol^{-1} cm^{-1}) = 10,000 L mol^{-1} cm^{-1}

l, path length of the sample (cm) = 1 cm

c = concentration (mol L^{-1})

Putting these values in the formula above, we get

$$0.02 = 10000 \times c \times 1$$

$$c = \frac{0.02}{10000} = 2 \times 10^{-6} \text{ mol} = 2 \mu M$$

Answer (2 μM)

28. Topic: CD, IR and NMR

- Infrared (IR) spectroscopy is used to identify functional groups within a molecule. Vibrating bonds in functional groups absorb energy at a frequency that corresponds to the vibrational frequency of the bond.
- Circular dichroism (CD) spectroscopy is a form of light absorption spectroscopy that measures the difference in absorbance of right- and left-circularly polarized light. CD spectroscopy is very sensitive to the secondary structure of polypeptides and is particularly powerful for monitoring conformational changes in the secondary structure of a protein.
- Nuclear magnetic resonance (NMR) is an analytical chemistry technique which is based upon the measurement of absorption of radiofrequency (RF) radiation by a nucleus in a strong magnetic field. It is used in quality control and research for determining the content and purity of a sample as well as its molecular structure.

Answer (C)

29. Topic: Electrophoresis

Polyacrylamide gel electrophoresis (PAGE) of proteins is typically carried out in polyacrylamide gels with a characteristic pore size, so the molecular separations are based on gel filtration (size and shape) as well as electrophoretic mobility (electric charge). The pH of the gel is high enough (pH = 12) so that both the peptides have net negative charges and move toward the positive electrode (anode) when the current is switched on.

Replacement of valine (whose pI = 6.03) by glutamate (whose pI = 5.78) in the hexapeptide Q will make its isoelectric point less than the pI of hexapeptide P. But, valine being molecular mass 99.1 D and glutamate being of molecular mass 129.1 D, will make P to move faster towards Q. It is because small molecules more easily fit through the pores in the gel, while larger ones have more difficulty.

Answer (A)

CHAPTER 7 Immunology

SYLLABUS

History of Immunology; Innate, Humoral and Cell Mediated Immunity; Antigen; Antibody Structure and Function; Molecular Basis of Antibody Diversity; Synthesis of Antibody and Secretion; Antigen-Antibody Reaction; Complement; Primary and Secondary Lymphoid Organs; B and T Cells and Macrophages; Major Histocompatibility Complex (MHC); Antigen Processing and Presentation; Polyclonal and Monoclonal Antibody; Regulation of Immune Response; Immune Tolerance; Hypersensitivity; Autoimmunity; Graft Versus Host Reaction.

CHAPTER ANALYSIS

Topic	GATE 2000	GATE 2011	GATE 2012	GATE 2013	GATE 2014	GATE 2015	GATE 2016	GATE 2017	GATE 2018	GATE 2019
History of Immunology										
Innate, Humoral and Cell Mediated Immunity	3								3	1
Antigen		1	1	1		1				
Antibody Structure and Function		2	1		1	1	1		1	
Molecular Basis of Antibody Diversity					1					
Synthesis of Antibody and Secretion										
Antigen–Antibody Reaction				1	1	2		1		
Complement			1		1	1	1			
Primary and Secondary Lymphoid Organ			1							
B and T Cells and Macrophages			1			1	1	1		
Major Histocompatibility Complex (MHC)	1	1		1					1	
Antigen Processing and Presentation								1		
Polyclonal and Monoclonal Antibody			1	1						
Regulation of Immune Response										
Immune Tolerance						1		1		
Hypersensitivity										
Autoimmunity								1		
Graft Versus Host Reaction										
Vaccination										
Plant Immunology					1					

QUESTIONS

1. Abzymes are
   (A) enzymes that are highly specific like antibodies.
   (B) antibodies that have catalytic activities.
   (C) also, referred to as zymogens.
   (D) enzymes which hydrolyze antibodies.

   (GATE 2000; 1 Mark)

2. Antibody diversity is generated by
   (A) Protein splicing.
   (B) Somatic mutation.
   (C) Allelic exclusion.
   (D) Inter-chromosomal recombination.

(GATE 2001; 1 Mark)

3. Recombinant live attenuated vaccine against hepatitis B was prepared from
   (A) plasma of chronically infected individual.
   (B) recombinant yeast expressing hepatitis B surface antigen.
   (C) recombinant vaccinia virus expressing hepatitis B surface antigen.
   (D) transgenic plants expressing hepatitis B surface antigen.

(GATE 2002; 1 Mark)

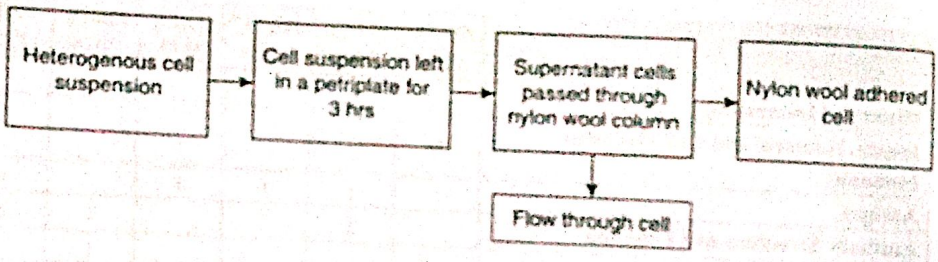
4. Viral replication within cells is inhibited by
   (A) IL4. (B) IL1.
   (C) IFNα. (D) TNFα.

(GATE 2002; 1 Mark)

5. Reverse vaccinology indicates
   (A) from antigenic protein to vaccine development.
   (B) from antigenic polysaccharide to vaccine development.
   (C) from antibody to vaccine development.
   (D) from genome sequence to vaccine development.

(GATE 2004; 2 Marks)

6. In a heterogeneous population of cells containing T cells, B cells and macrophages, the cells are separated in the following scheme



Identify the major population of cells present in petriplate, nylon wool adhered and nylon wool column flow through respectively.
   (A) Macrophage, B cell, T cell
   (B) T cell, B cell, macrophage
   (C) Macrophage, T cell, B cell
   (D) B cell, T cell, macrophage

(GATE 2004; 2 Marks)

7. Determine the correctness or otherwise of the following Assertion (*A*) and Reason (*R*).

Assertion: An antigen recognized by one immunoglobulin subtype is not recognized by any other subtype.

Reason: Immunoglobulin subtypes differ from each other both in the variable and in the constant regions.

   (A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
   (B) Both (*A*) and (*R*) are true but (*R*) is not the correct reason for (*A*).
   (C) Both (*A*) and (*R*) are false.
   (D) (*A*) is true but (*R*) is false.

(GATE 2005; 2 Marks)

8. Match the items Group I with Group II.

Group I	Group II
P. Programmed cell death at site of infection	1. TMV coat protein
Q. Hormone upregulated during flooding stress	2. EPSP synthase
R. Target for herbicide glyphosate	3. Hypersensitive response
S. Pathogen-derived resistance	4. Ethylene

	P	Q	R	S
(A)	1	2	4	3
(B)	3	4	2	1
(C)	1	4	2	3
(D)	3	2	4	1

(GATE 2005; 2 Marks)

9. Which of these mice fail to develop a thymus?
   (A) Motheaten mice (B) Beige mice
   (C) Knock out mice (D) Nude mice

(GATE 2006; 2 Marks)

10. Match items in Group I with correct options from those given in Group II.

Group I	Group II
P. Tissue plasminogen activator	1. Immunomodulator
Q. Gamma interferon	2. Biodegradable plastic
R. Podophyllotoxin	3. Clot dissolution
S. Polyhydroxyalkanoate	4. Anti-tumor agent

	P	Q	R	S
(A)	4	3	1	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	3	1	4	2

(GATE 2006; 2 Marks)

Statement for Linked Answer Questions 11–12:

Absence of cellulosic cell wall, high β-carotene content and GRAS status make *Dunaliella salina* a good model system for producing edible vaccines. 10^9 Cells of *D. salina* were electroporated with a high expression DNA vector containing an antigenic gene.

11. If 10^3 cells survived after electroporation, how many cells were killed during this process (round of to the nearest number)?

(A) 10^9 (B) 10^8
(C) 10^6 (D) 10^5

(GATE 2007; 2 Marks)

12. The antigen is expressed as transmembrane protein with a single epitope on its extracellular domain. The cells that survived (assume 100% transfection and expression of protein) were incubated with a radiolabeled Fab fragment (specific activity: 100 cpm picomole^{-1}) against this epitope. After washing, the cell pellet has 1000 cpm. The average number of epitopes present on a single recombinant alga are

(A) 6×10^9 (B) 1×10^9
(C) 6×10^3 (D) 1×10^6

(GATE 2007; 2 Marks)

13. Interferon β is produced by

(A) bacteria infected cells.
(B) virus infected cells.
(C) both virus and bacteria infected cells.
(D) fungi infected cells.

(GATE 2010; 1 Mark)

14. A neonatally thymectomized mouse, immunized with protein antigen shows

(A) both primary and secondary responses to the antigen.
(B) only primary response to the antigen.
(C) delayed type hypersensitive reactions.
(D) no response to the antigen.

(GATE 2010; 1 Mark)

15. Lymphocytes interact with foreign antigens in

(A) bone marrow.
(B) peripheral blood.
(C) thymus.
(D) lymph nodes.

(GATE 2010; 1 Mark)

16. Dizygotic twins are connected to a single placenta during their embryonic development. These twins

(A) have identical MHC haplotypes.
(B) have identical T_H cells.
(C) have identical T cells.
(D) can accept grafts from each other (both (A) and (B)).

(GATE 2010; 2 Marks)

17. Members of the antibody protein family that have common structural features are collectively known as

(A) haptens.
(B) allergens.
(C) antigens.
(D) immunoglobulins.

(GATE 2011; 1 Mark)

18. In ABO blood group system, antigenic determinants are

(A) nucleic acid. (B) carbohydrate.
(C) lipid. (D) protein.

(GATE 2011; 1 Mark)

19. The polymorphic domains for Class II MHC proteins are

(A) α_1 and β_2 domains only.
(B) β_1 and α_2 domains only.
(C) α_1 and β_1 domains only.
(D) α_2 and β_2 domains only.

(GATE 2011; 1 Mark)

20. Determine the correctness or otherwise of the following Assertion (*A*) and the Reason (*R*).

Assertion: IgM is found in serum as a pentameric protein consisting of five IgM monomers.

Reason: The pentameric form of IgM is due to cross-linking of IgM monomers via peptide bond.

(A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(B) Both (*A*) and (*R*) are true but (*R*) is not the correct reason for (*A*).
(C) (*A*) is true but (*R*) is false.
(D) (*A*) is false but (*R*) is true.

(GATE 2011; 2 Marks)

21. Idiotypic determinants of an antibody are associated with the
(A) constant region of the heavy chains.
(B) constant region of the light chains.
(C) variable region.
(D) constant regions of light and heavy chains.

(GATE 2012; 1 Mark)

22. Identification of blood groups involves
(A) precipitation. (B) neutralization.
(C) opsonization. (D) agglutination.

(GATE 2012; 1 Mark)

23. B-lymphocytes originate from the bone marrow, whereas T lymphocytes originate from
(A) thymus. (B) bone marrow.
(C) spleen. (D) liver.

(GATE 2012; 1 Mark)

24. A humanized antibody is one in which the
(A) heavy and light chains are from human.
(B) heavy chain is from human and light chain is from mouse.
(C) light chain is from human and heavy chain is from mouse.
(D) CDRs are from mouse, and the rest is from human.

(GATE 2012; 1 Mark)

25. Nude mice refers to
(A) mice without skin.
(B) mice without thymus.
(C) knockout mice.
(D) transgenic mice.

(GATE 2012; 1 Mark)

26. Heat inactivation of serum is done to inactivate
(A) prions.
(B) mycoplasma.
(C) complement.
(D) pathogenic bacteria.

(GATE 2012; 1 Mark)

27. Protein A, which has strong affinity to Fc region of immunoglobulin, is extracted from
(A) *Saccharomyces cerevisiae.*
(B) *Staphylococcus aureus.*
(C) *Streptococcus pyogenes.*
(D) *Streptococcus sanguis.*

(GATE 2013; 1 Mark)

28. The first humanized monoclonal antibody approved for the treatment of breast cancer is
(A) rituximab. (B) cetuximab
(C) bevacizumab. (D) herceptin.

(GATE 2013; 1 Mark)

29. The role of an adjuvant is to
(A) prolong the persistence of antigen.
(B) cross link the antigen.
(C) increase the size of antigen.
(D) avoid inflammation.

(GATE 2013; 1 Mark)

30. Endogenous antigens are presented on to the cell surface along with
(A) MHC II.
(B) MHC I.
(C) $F_c\gamma$ receptor.
(D) Complement receptor.

(GATE 2013; 1 Mark)

31. Gram-positive bacteria are generally resistant to complement-mediated lysis because
(A) thick peptidoglycan layer prevents insertion of membrane attack complex into the inner membrane.
(B) Gram-positive bacteria import the membrane attack complex and inactivate it.
(C) membrane attack complex is degraded by the proteases produced by the Gram-positive bacteria.
(D) Gram-positive bacteria cannot activate the complement pathway.

(GATE 2014; 1 Mark)

32. Prior exposure of plants to pathogens is known to increase resistance to future pathogen attacks. This phenomenon is known as
(A) systemic acquired resistance.
(B) hypersensitive response.
(C) innate immunity.
(D) antibody mediated response

(GATE 2014; 1 Mark)

33. Reactions between antibodies and antigens that are detected by precipitate formation in an agar gel are referred as

(A) immunoprecipitation assay.
(B) immunodiffusion assay.
(C) immunoaggregation assay.
(D) immunofixation assay.

(GATE 2014; 1 Mark)

34. In a mouse genome, the numbers of functional V_α, J_α, V_β, D_β, J_β gene segments are 79, 38, 21, 2 and 11, respectively. The total number of possible combinations for $\alpha\beta$-T cell receptors are _____ $\times 10^6$.

(GATE 2014; 2 Marks)

35. Match the immunoglobulin class in Group I with its properties in Group II.

Group I	Group II
P. IgG	1. Major antibody in external secretions such as bronchial mucus
Q. IgA	2. Protects against parasites
R. IgE	3. Antibody that appears first in serum after exposure to an antigen
S. IgM	4. Antibody present in highest concentration in serum

(A) P–4; Q–1; R–2; S–3 (B) P–3; Q–2; R–1; S–4
(C) P–4; Q–3; R–1; S–2 (D) P–1; Q–4; R–3; S–2

(GATE 2014; 2 Marks)

36. Which one of the following complement proteins is the initiator of the membrane attack complex?

(A) C3a (B) C3b
(C) C5a (D) C5b

(GATE 2015; 1 Mark)

37. Based on the heavy chain, which one of the following antibodies has multiple subtypes?

(A) IgM (B) IgD
(C) IgE (D) IgG

(GATE 2015; 1 Mark)

38. Anergy refers to

(A) mitochondrial dysfunction.
(B) allergy to environmental antigens.
(C) unresponsiveness to antigens.
(D) a state of no energy.

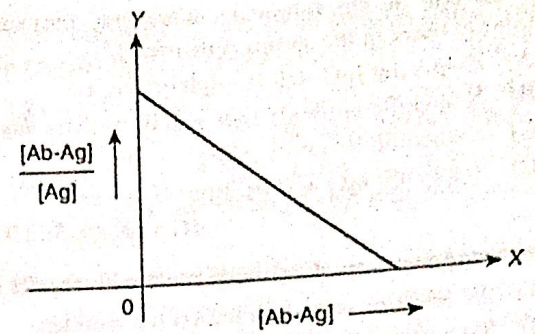
(GATE 2015; 1 Mark)

39. ABO blood group antigens in humans are differentiated from each other on the basis of

(A) sialic acid. (B) lipids.
(C) spectrin. (D) glycoproteins.

(GATE 2015; 1 Mark)

40. The interaction between an antigen (Ag) and a single-chain antibody (Ab) was studied using Scatchard analysis. The result is shown below.



The affinity of interaction and the total concentration of antibody, respectively, can be determined from

(A) slope and y-intercept. (B) y-intercept and slope.
(C) x-intercept and slope. (D) slope and x-intercept

(GATE 2015; 1 Mark)

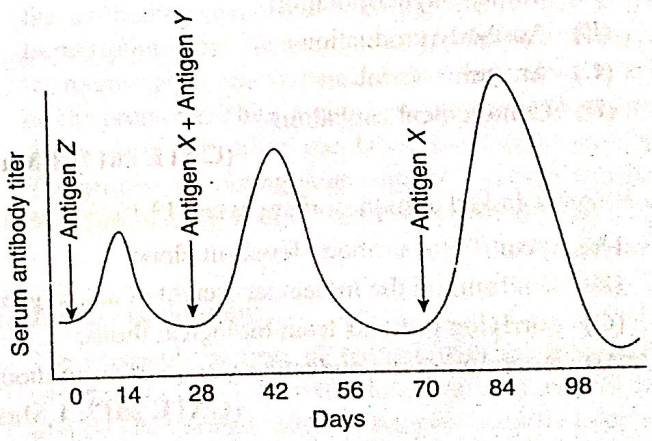
41. Match the cells in Group I with their corresponding entries in Group II.

Group I	Group II
P. Mast cells	1. Activation of the complement pathway
Q. Natural killer cells	2. Expression of CD56
R. Neutrophils	3. Contains azurophilic granules
S. Dendritic cells	4. Defense against helminthic infection
	5. Production of antibodies specific to bacteria
	6. Contains long membranous projections

(A) P–4; Q–2; R–3; S–5 (B) P–4; Q–2; R–3; S–6
(C) P–3; Q–1; R–2; S–5 (D) P–3; Q–1; R–2; S–6

(GATE 2015; 2 Marks)

42. Three distinct antigens X, Y and Z were used to raise antibodies. Antigen Z was injected in a mouse on day zero followed by the administration of antigens X and Y on day 28. A second injection of antigen X was administered on day 70. The antibody titers were monitored in the serum every day and the results are shown below:



Which one of the following statements regarding the antibody titers in the serum is incorrect?

(A) Z-specific IgG will be high on day 14
(B) X-specific antibody titer will be high on day 84
(C) X-specific IgG will be high on day 42
(D) Y-specific IgG will be high on day 84

(GATE 2015; 2 Marks)

43. Junctional diversity of antibody molecules results from

(A) the addition of switch region nucleotides.
(B) the addition of N and P nucleotides.
(C) the joining of V, D and J segments.
(D) mutations in complementarity-determining regions.

(GATE 2016; 1 Mark)

44. Find the incorrect combination.

(A) Surface immunoglobulins – B cell antigen receptor
(B) Affinity maturation – Isotype switching
(C) F_c region of antibodies – Binding to complement proteins
(D) Spleen, the secondary lymphoid organ – No connection with the lymphatic system

(GATE 2016; 2 Marks)

45. Which of the following statement(s) is/are CORRECT for antigen activated effector T cells?

P. CD4+ cells make contact with macrophages and stimulate their microbicidal activity.
Q. CD4+ cells make contact with B cells and stimulate them to differentiate into plasma cells.
R. CD8+ cells make contact with B cells and stimulate them to differentiate into plasma cells.
S. CD8+ cells make contact with virus infected cells and kill them.

(A) Q only
(B) Q and S only
(C) P, Q and S only
(D) P, Q, R and S

(GATE 2016; 2 Marks)

46. Which one of the following mechanisms is used by human pathogens to evade host immune responses?

(A) Somatic hypermutation
(B) Antibody production
(C) Antigenic variation
(D) Complement activation

(GATE 2017; 1 Mark)

47. Enzyme-linked immunosorbent assay (ELISA) is used for

(A) quantifying antibody levels in blood.
(B) determining the molecular weight of an antigen.
(C) purifying proteins from biological fluids.
(D) determining the molecular weight of an antibody.

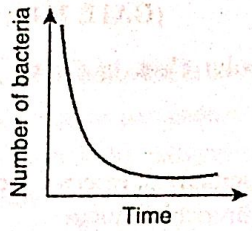
(GATE 2017; 1 Mark)

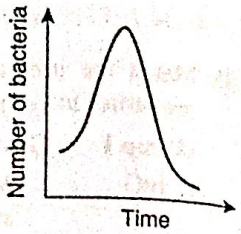
48. Macrophages eliminate pathogenic bacteria upon activation by

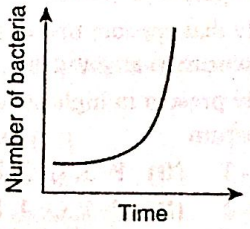
(A) NK cells.
(B) basophils.
(C) CD4+ T cells.
(D) plasma cells.

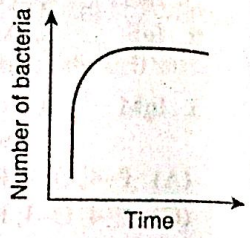
(GATE 2017; 1 Mark)

49. An immunocompetent person becomes infected with a pathogenic strain of bacteria. Which one of the following graphs correctly depicts bacterial load in this person over time?

(A)


(B)


(C)


(D)


(GATE 2017; 2 Marks)

50. Which of the following conditions promote the development of human autoimmune disorders?

P. Inability to eliminate self-reactive lymphocytes
Q. Generation of auto-antibodies
R. Ability to eliminate self-reactive T cells
S. Induction of regulatory T cells in the thymus

(A) P, R
(B) Q, S
(C) P, Q
(D) R, S

(GATE 2017; 2 Marks)

51. Which one of the following is not a principal component of innate immunity?

(A) Mucosal epithelia
(B) Dendritic cells
(C) Complement system
(D) Memory B cells

(GATE 2018; 1 Mark)

52. Which of the following cytokines are endogenous pyrogens?

P. Tumor necrosis factor α
Q. Interleukin 1
R. Transforming growth factor β
S. Interleukin 10

(A) P and Q only
(B) P and R only
(C) R and S only
(D) Q and S only

(GATE 2018; 2 Marks)

53. Which one of the following amino acids is catalyzed by activated macrophages to produce reactive nitrogen species?

(A) Arginine (B) Asparagine
(C) Cysteine (D) Histidine

(GATE 2018; 2 Marks)

54. Which one of the combinations of the following statements is true about antibody structure?

P. Limited proteolysis of rabbit IgG with the enzyme pepsin generates two antigen-binding regions (Fab) and an Fc fragment.

Q. Limited proteolysis of rabbit IgG with the enzyme papain generates a single bivalent antigen-binding region $F(ab')_2$ and peptide fragments.

R. The Fc fragment of IgG can self-associate and crystallize into a lattice.

S. The F(ab′)2 fragment of IgG is composed of both light and heavy chains.

(A) P and Q only (B) P and R only
(C) R and S only (D) Q and S only

(GATE 2018; 2 Marks)

55. Which one of the following statements is true with regard to processing and presentation of protein antigens?

(A) In the class II MHC pathway, protein antigens in the cytosol are processed by proteasomes.

(B) In the class I MHC pathway, extracellular protein antigens are endocytosed into vesicles and processed.

(C) In the class I MHC pathway, transporter associated antigen processing (TAP) protein is required for translocating processed peptides generated in the cytosol.

(D) Invariant chain in endoplasmic reticulum is involved in transport of peptides and loading of class I MHC.

(GATE 2018; 2 Marks)

56. Which one of the following is NOT a part of the human non-specific defense system?

(A) Interferon (B) Mucous
(C) Saliva (D) Antibody

(GATE 2019; 1 Mark)

ANSWER KEY

1. (B)	2. (D)	3. (B)	4. (C)	5. (D)	6. (A)	7. (C)	8. (B)	9. (D)	10. (D)
11. (A)	12. (A)	13. (B)	14. (B)	15. (D)	16. (A)	17. (D)	18. (B)	19. (C)	20. (C)
21. (C)	22. (D)	23. (B)	24. (D)	25. (B)	26. (C)	27. (B)	28. (D)	29. (A)	30. (B)
31. (A)	32. (A)	33. (B)	34. (1.38×10^6)		35. (A)	36. (D)	37. (D)	38. (C)	39. (D)
40. (D)	41. (B)	42. (D)	43. (B)	44. (B), (D)	45. (C)	46. (C)	47. (A)	48. (C)	49. (B)
50. (C)	51. (D)	52. (A)	53. (A)	54. (C)	55. (C)	56. (D)			

ANSWERS WITH EXPLANATIONS

1. **Topic: Polyclonal and Monoclonal Antibody**

An abzyme is a monoclonal antibody with catalytic properties. It is also called catmab that is catalytic monoclonal antibody. They are potential tools in biotechnology, e.g., they perform specific actions on DNA.

Answer (B)

2. **Topic: Molecular Basis of Antibody Diversity**

Each type of antibody chain—κ light chains, λ light chains, and heavy chains—has a separate pool of gene segments and exons from which a single polypeptide chain is eventually synthesized. Each pool is on a different chromosome and contains a large number of gene segments encoding the V region of an antibody chain and, a smaller number of exons encoding the C region.

During the development of a B cell, a complete coding sequence for each of the two antibody chains to be synthesized is assembled by site-specific genetic recombination. In addition to bringing together the separate gene segments and the C-region exons of the antibody gene, these rearrangements also activate transcription from the gene promoter through changes in the relative positions of the enhancers and silencers acting on the promoter. Thus, a complete antibody chain can be synthesized only after the DNA has been rearranged. The process of joining gene segments contributes to the diversity of antigen-binding sites in several ways.

Answer (D)

3. **Topic: Vaccination**

The classical example of recombinant protein vaccines currently in use in humans is the vaccine against hepatitis B. The current vaccines are produced by expressing the hepatitis B surface antigen (HBsAg) in yeast cells.

The HBsAg assembles into virus-like particles (VLPs), which are extremely immunogenic, making the HBV vaccine a very efficacious vaccine. The yeast expression system may secrete the antigen into the culture supernatant that can facilitate its purification. Furthermore, yeast cells offer some of the eukaryotic cellular machinery responsible for the post-translational modification of proteins, being capable of rendering proteins glycosylated.

Answer (B)

4. Topic: Innate, Humoral and Cell Mediated Immunity

IFNα and IFNβ can induce an antiviral response or resistance to viral replication by binding to the IFN α/β receptor. Once bound, IFNα and IFNβ activate the JAK-STAT pathway, which in turn induces the transcription of several genes. One of these genes encodes an enzyme known as 2′-5′-oligo-adenylate synthetase [2-5(A) synthetase], which activates a ribonuclease (RNAse L) that degrades viral RNA. Other genes activated by IFN α/β binding to its receptor also contribute to the inhibition of viral replication. For example, IFN α/β binding induces a specific protein kinase called dsRNA-dependent protein kinase (PKR), which inactivates protein synthesis, thus blocking viral replication in infected cells.

Answer (C)

5. Topic: Vaccination

Reverse vaccinology is the name given to the examination of the genome of an organism in order to identify novel antigens and epitopes that might constitute vaccine candidates. Current reverse vaccinology approaches include comparative in silico analyses of multiple genome sequences in order to identify conserved antigens within a heterogeneous pathogen population and identification of antigens that are unique to pathogenic isolates but not present in commensal strains. In addition, transcriptomic and proteomic data sets are integrated into a selection process that yields a short list of candidate antigens to be tested in animal models, thus reducing the costs and time of downstream analyses.

Answer (D)

6. Topic: B and T Cells and Macrophages

Cells may be separated by virtue of their affinity to fibers and solid surfaces such as plastic Petri dishes. The use of this method has largely been restricted to the separation of lymphoid cells which have a natural affinity for nylon or glass wool. The method is based on the fact that T cells do not adhere to nylon, whereas other lymphoid cells such as B cells, monocytes/macrophages and neutrophils do adhere. Column of unmodified fibers preferentially bind the B cells, monocytes/macrophages and neutrophils allowing the T cells to pass straight through. Macrophages attach to petri plates and thus, its major population will be found on petri plates.

Answer (A)

7. Topic: Antibody Structure and Function

Antibodies are classified into five main classes or isotypes — IgA, IgD, IgE, IgG and IgM. They are classed according to the heavy chain they contain—alpha, delta, epsilon, gamma or mu respectively. These differ in the sequence and number of constant domains, hinge structure and the valency of the antibody. Although these are relatively dissimilar in protein sequence they share a similar structure and function. Immunoglobulin-like domains that are related in both sequence and structure can be found in several diverse protein families. Thus, it is possible for an antigen to be recognized by more than one immunoglobulin subtype.

Answer (C)

8. Topic: Plant Immunology

- The hypersensitive response is induced by certain plant pathogens and involves programmed cell death to restrict the spread of pathogens from the infection site. Concurrent with the induction of cell death, the host activates a defense response.
- Internal changes in oxygen and ethylene are considered primary signals triggering plant-adaptive responses to flooding. All cells of a plant are capable of synthesizing ethylene, and endogenous levels are largely determined by biosynthesis rates and amounts lost by diffusion to the external environment.
- EPSP synthase is the site of action of the commercially important herbicide glyphosate, to which *Chlamydomonas* is sensitive. Glyphosate binds to a pocket within the EPSPS protein that normally binds phosphoenol pyruvate. The binding of glyphosate to the phosphoenol pyruvate site mimics an intermediate state of the ternary enzyme–substrate complex. By preventing the conversion of the substrate shikimate 3-phosphate to 5-enolpyruvylshikimate-3-phosphate, the plant is unable to produce aromatic amino acids required for survival.
- Genes that confer viral pathogen-derived resistance (PDR) include those for coat proteins, replicases, movement proteins, defective interfering RNAs and DNAs, and non-translated RNAs.

Answer (B)

9. Topic: Primary and Secondary Lymphoid Organ

Nude mice have no thymus and therefore no T cells, a class of lymphocytes that depend on the thymus to develop. For lack of T cells, nude mice cannot reject tumors or transplants of cells from humans or other animals.

Answer (D)

10. Topic: Innate, Humoral and Cell-Mediated Immunity

- Tissue plasminogen activator (tPA) is a serine protease found in vascular endothelial cells. It is involved in maintaining balance between thrombolysis and thrombogenesis. It induces a conformational change that catalyzes the conversion of plasminogen to plasmin.
- Interferon gamma (IFN-γ) is a cytokine critical to both innate and adaptive immunity, and functions as the primary activator of macrophages, in addition to stimulating natural killer cells and neutrophils. It is a pleotropic cytokine that induces antiviral, antiproliferative, and immunomodulatory effects on numerous target cells.
- Podophyllotoxin (PTOX) is an important plant-derived natural product, has derivatives such as etoposide and teniposide, which have been used as therapies for cancers and venereal wart. PTOX structure is closely related to the aryltetralin lactone lignans that have antineoplastic and antiviral activities.
- Polyhydroxyalkanoates (PHAs) are naturally produced polyesters that are synthesized and intracellularly accumulated as a carbon and energy storage material by various microorganisms. They include poly-3-hydroxybutyrate (PHB or PH3B), PHV, and PHH. A PHA copolymer called poly (3-hydroxybutyrate-co-3-hydroxyvalerate) (PHBV) is less stiff and tougher, and it may be used as packaging material.

Answer (D)

11. Topic: Antigen Processing and Presentation

Number of killed cells can be calculated by subtracting number of cells survived from total number of cells.

Number of cells killed = Total number of cells – Number of cells survived

$$= 10^9 - 10^3$$
$$= 9.9 \times 10^8 \sim 10^9$$

Answer (A)

12. Topic: Antigen Processing and Presentation

The cell pellet has 10^3 cpm. It is due to Fab binding to epitope present on recombinant living cells per algae.

$$\text{Total Fab or epitope} = \frac{\text{Radioactivity in pellet}}{\text{Specific activity}}$$

$$= \frac{10^3 \text{ cpm}}{10^2 \text{ cpm picomole}^{-1}}$$

$$= \frac{10^3 \times 10^{-12}}{10^2} \text{ mole}$$

$$= 10 \times 10^{-12} \text{ mole}$$

$$= 10 \text{ picomole}$$

Thus, 1000 cells have 10 picomole epitopes, if we assume transfection and expression of protein to be 100%.

Therefore, average number of epitopes on single recombinant alga would be

$$= \frac{10 \times 10^{-12}}{10^3} \text{ mole}$$

$$= 10^{-14} \text{ mole} = 10^{-14} \times 6.022 \times 10^{23} = 6 \times 10^9$$

Answer (A)

13. Topic: Cell Mediated Immunity

Interferon comprises a group of proteins produced by virus-infected cells. Interferons increases in the expression of class I and class II MHC molecules, and the augment NK-cell activity. Among the many functions of the interferons is the ability to bind to nearby cells and induce a generalized antiviral state. They can be of three types: α, β and γ.

Answer (B)

14. Topic: Humoral Immunity

In neonatal thymectomy, the thymus is surgically removed from newborn mice. These thymectomized mice show a dramatic decrease in circulating lymphocytes of the T cell lineage and an absence of cell-mediated immunity. The initial encounter of a naive immunocompetent lymphocyte with a protein antigen induces a primary response with B lymphocyte.

Answer (B)

15. Topic: Cell Mediated Immunity

When a foreign antigen gains entrance to the tissues, it is picked up by the lymphatic system (which drains all the tissues of the body) and is carried to various organized lymphoid tissues such as lymph nodes, which trap the foreign antigen. As lymph passes from the tissues to lymphatic vessels, it becomes progressively enriched in lymphocytes. Thus, the lymphatic system also serves as a means of transporting lymphocytes and antigen from the connective tissues to organized lymphoid tissues where the lymphocytes may interact with the trapped antigen and undergo activation.

Answer (D)

16. Topic: Major Histocompatibility Complex (MHC)

Dizygotic twins, also called fraternal twins or non-identical twins, two siblings who come from separate ova, or eggs, that are released at the same time from an ovary and are fertilized by separate sperm. Monozygotic or identical twins share same placenta, and dizygotic twins normally have two separate placentae. But few dizygotic twins that share single placenta are likely to have identical HLA haplotypes more often than ordinary siblings and thus might be genetically more alike.

Answer (A)

17. Topic: Antibody Structure and Function

Immunoglobulins or antibodies are globular proteins produced by plasma cells in order to defense our body against an immunogen. Each antibody or immunoglobulin is a Y-shaped molecule made up of four polypeptide chains, 2 light chains and 2 heavy chains, i.e., H_2L_2. The amino terminal ends of the polypeptide chains show considerable variation in amino acid composition and are referred to as the variable (V) regions to distinguish them from the relatively constant (C) regions.

Answer (D)

18. Topic: Antigen

In ABO blood group system, antigens are primarily determined by carbohydrates residues. Antigens are composed of fructose, galactose, and N acetyl-galactosamine or N acetyl-glucosamine. N acetyl-galactosamine is present in antigen A and N acetyl-glucosamine is present in antigen B.

Answer (B)

19. Topic: Major Histocompatibility Complex (MHC)

Class II MHC molecules belong to class of major histocompatibility complex molecules usually present on the surface of the antigen presenting cells such as dendritic cells, mononuclear phagocytes etc. These molecules are heterodimers consists of α and β chain. The sub-designations used are α_1 α_2 β_1 β_2, etc. α_1 and β_1 form the polymorphic domain which form membrane-distal peptide-binding domain and α_2 and β_2 regions form the remaining extracellular parts of the chains.

Answer (C)

20. Topic: Antibody Structure and Function

IgM is the largest pentameric antibody, and it is the most recent antibody to appear in the response to exposure to an antigen. IgM present in serum as a pentamer of disulfide-linked immunoglobulin molecules joined by a single cross-linking peptide (J-chain).

Answer (C)

21. Topic: Antibody Structure and Function

Idiotypic determinants of an antibody are associated with its variable region, i.e., antigen-binding site of the antibody. The idiotopes can further divide into three categories depending on its position; alpha-idiotopes, beta-idiotopes and gamma-idiotopes.

Answer (C)

22. Topic: Antigen

Identification blood group depends upon agglutination reaction takes place between antibody and antigen present on cell surface of red blood cells. Agglutination refers to the clumping of cells or particles. It comes from Latin word "agglutinare" which means "glueing to".

Answer (D)

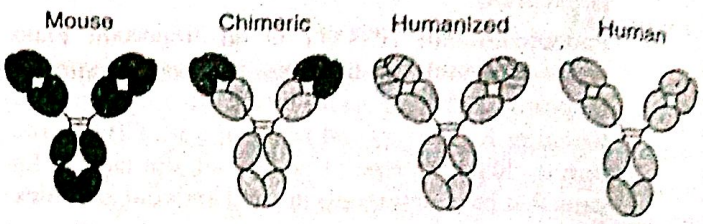
23. Topic: B and T Cells and Macrophages

All lymphocytes are originated from common lymphocyte progenitor cells in bone marrow. B-lymphocytes originate as well as mature in bone marrow (or bursa of fabricius in birds) however T-lymphocytes migrates towards thymus from bone marrow for maturation.

Answer (B)

24. Topic: Polyclonal and Monoclonal Antibody

A humanized antibody refers to the non-human species antibody in which protein sequences are modified to increase the similarities between it and antibody produced naturally by the human.



Answer (D)

25. Topic: Primary and Secondary Lymphoid Organs

Nude mice are the laboratory mice grown with a genetic mutation, which results in deteriorated or absent thymus in them. Lack or deteriorated thymus gives these mice a phenotype of absence of hairs on skin, and these mice appear as nude.

Answer (B)

26. Topic: Complement

Heat inactivation of serum is done to inactivate complement. Complement is the group of proteins involved in immune system. These are inactivated only in tests in which they could influence the results such as virus assay, cytotoxicity assays, etc.

Answer (C)

27. Topic: Antigen-Antibody Reaction

Staphylococcus aureus is a round shaped gram-positive bacteria which has a special protein (protein A) in cell its cell wall. This protein has very high affinity towards Fc region of immunoglobulin due to which they are widely used as affinity matrices IgG purification.

Answer (B)

28. Topic: Polyclonal and Monoclonal Antibody

- Rituximab (trade name Rituxan, MabThera, etc.) are used to treat few autoimmune diseases and cancers such as chronic lymphocytic leukemia, rheumatoid arthritis, granulomatosis with polyangiitis etc.

- Cetuximab (trade name EGF receptor) used to treat metastatic colorectal cancer, lung cancer and head and neck cancer.
- Bevacizumab (trade name Avastin) is used to treat various cancer and specific eye disease.
- Trastuzumab (trade name Herceptin) is a humanized monoclonal antibody obtained from mouse used to treat breast cancer.

Answer (D)

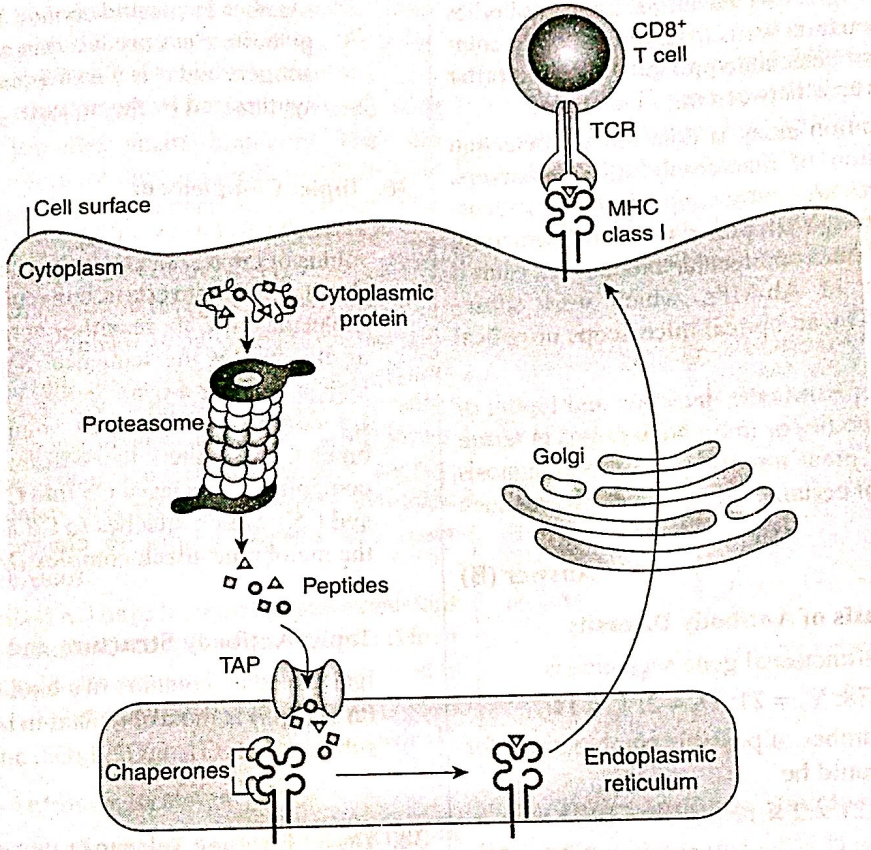
29. Topic: Antigen

Adjuvant refers to the substance used to increase the body's immune response to a particular antigen. It prevents the immediate removal of antigen from the body and thus provides enough time for affinity maturation of large amounts of antibodies.

Answer (A)

30. Topic: Major Histocompatibility Complex (MHC)

MHC I majorly participate in presenting endogenous protein or antigens on to the cell surface. Endogenous antigens are first digested in proteasome after which the peptides get associated with MHC I in endoplasmic reticulum. MHC I-peptide complex then passes through Golgi for further maturation and processing and finally passes towards plasma membrane, where MHC I presents antigenic peptides to cytoplasmic T lymphocyte cells.



Answer (B)

31. Topic: Complement

Gram-positive bacteria are generally resistant to complement-mediated lysis because the thick peptidoglycan layer in their cell wall prevents insertion of the MAC into the inner membrane. Although complement activation can occur on the cell membrane of encapsulated bacteria such as *Streptococcus pneumoniae*, the capsule prevents interaction between C3b deposited on the membrane and the CR1 on phagocytic cells. Some bacteria possess an elastase that inactivates C3a and C5a, preventing these split products from inducing an inflammatory response.

Answer (A)

32. Topic: Plant Immunology

Upon infection with necrotizing pathogens many plants develop an enhanced resistance to further pathogen attack also in the uninoculated organs. This type of enhanced resistance is referred to as systemic acquired resistance (SAR). In the SAR state, plants are primed (sensitized) to more quickly and more effectively activate defense responses the second time they encounter pathogen attack. Since SAR depends on the ability to access past experience, acquired disease resistance is a paradigm for the existence of a form of plant memory.

Answer (A)

33. Topic: Antigen-Antibody Reaction

- Immunoprecipitation test can be used to detect antibodies or antigens. This test is based on a precipitation reaction in which antibodies called precipitins react with antigens, diffuse toward each other, and form a visible precipitate. Lattice like networks of these complexes, which are visually opaque, form minutes to hours later.
- Immunodiffusion tests are based on the same principle as the precipitin test, but they are carried out in a thin layer of agar that has solidified on a glass slide. These tests are used to determine if more than one antigen—and therefore more than one antibody—is present in a serum specimen. Small wells are made in the solidified agar, and the antigens and antibodies are placed in separate wells. Antigen-antibody complexes appear as detectable precipitation lines (after staining) in the agar between the wells.
- Immunoaggregation assay is done for the detection and quantification of macromolecular biomarkers. Antibodies (Abs) are firstly conjugated to microparticles (MPs) through streptavidin-biotin interaction; the addition of macromolecular biomarkers caused the aggregation of Ab-MPs, which were subsequently detected by an optical microscope or optical particle sizer.
- Immunofixation permits the detection and typing of monoclonal antibodies or immunoglobulins in serum or urine. It is of great importance for the diagnosis and monitoring of certain blood related diseases such as myeloma.

Answer (B)

34. Topic: Molecular Basis of Antibody Diversity

Given, the number of functional gene segments is

$$V_\alpha = 79;\ J_\alpha = 38;\ V_\beta = 21;\ D_\beta = 2;\ J_\beta = 11$$

Therefore, the total number of possible combinations for $\alpha\beta$-T cell receptors would be

$$= 79 \times 38 \times 21 \times 2 \times 11 = 1.38 \times 10^6$$

Answer (1.38×10^6)

35. Topic: Antibody Structure and Function

- IgG is the most abundant class in serum, constitutes about 80% of the total serum immunoglobulin. The IgG molecule consists of two γ heavy chains and two κ or two λ light chains.
- Although IgA constitutes only 10–15% of the total immunoglobulin in serum, it is the predominant immunoglobulin class in external secretions such as breast milk, saliva, tears, and mucus of the bronchial, genitourinary, and digestive tracts. In serum, IgA exists primarily as a monomer, but polymeric forms (dimers, trimers, and some tetramers) are sometimes seen in external secretions, called secretory IgA.
- The potent biological activity of IgE allowed it to be identified in serum despite its extremely low average serum concentration (0.3 $\mu g\ mL^{-1}$). IgE antibodies mediate the immediate hypersensitivity reactions that are responsible for the symptoms of hay fever, asthma, hives, and anaphylactic shock.
- IgM accounts for 5–10% of the total serum immunoglobulin, with an average serum concentration of 1.5 $mg\ mL^{-1}$. Monomeric IgM, with a molecular weight of 180,000, is expressed as membrane-bound antibody on B cells. IgM is secreted by plasma cells as a pentamer in which five monomer units are held together by disulfide bonds. IgM is the first immunoglobulin class produced in a primary response to an antigen, and it is also the first immunoglobulin to be synthesized by the neonate.

Answer (A)

36. Topic: Complement

Hydrolysis of a short fragment (C3a) from the amino terminus of the α chain by the C3 convertase generates C3b. A single C3 convertase molecule can generate over 200 molecules of C3b, resulting in tremendous amplification at this step of the sequence. Some of the C3b binds to C4b2a to form a trimolecular complex C4b2a3b, called C5 convertase. The C3b component of this complex binds C5 and alters its conformation, so that the C4b2a component can cleave C5 into C5a, which diffuses away, and C5b, which attaches to C6 and initiates formation of the membrane attack complex (MAC).

Answer (D)

37. Topic: Antibody Structure and Function

IgG molecule contains two binding sites. Immunoglobulin G (IgG) is most abundant in human serum. It has four subclasses, IgG1, IgG2, IgG3, and IgG4.

Answer (D)

38. Topic: Immune Tolerance

Anergy is a state of immune unresponsiveness. It is induced when the T cell's antigen receptor is stimulated, effectively freezing T cell responses pending a "second signal" from the antigen-presenting cell.

Answer (C)

39. Topic: Antigen

The surfaces of erythrocytes contain a genetically determined assortment of antigens composed of glycoproteins and glycolipids. Based on the presence or absence of various antigens, blood is categorized into different blood groups.

Answer (D)

40. Topic: Antigen-Antibody Reaction

Scatchard plots are built between Ab-Ag/Ag versus Ab-Ag. In these plots, Ab-Ag equals moles of bound antigen/mole antibody and Ag is the concentration of free antigen. From a Scatchard plot, both the equilibrium constant (K_a) and the number of binding sites per antibody molecule (n), or its valency, can be obtained. If all antibodies have the same affinity, then a Scatchard plot yields a straight line with a slope of $-K_a$. The x intercept is n, the valency of the antibody.

Answer (D)

41. Topic: B and T Cells and Macrophages

- Mast cells are present in connective tissue and alongside blood vessels. They are responsible for releasing histamine and are associated with allergies. They have been suggested to be major effector cells in the immune response to infection with helminths.
- Natural killer (NK) cells kill cells infected with viruses and provides innate immunity. They are called natural because they are ready to kill as soon as they are formed. After attaching to foreign cells, they release enzymes and other substances that damage the outer membranes of the foreign cells. CD56 is a pathogen recognition receptor on these cells.
- Neutrophils, also known as polymorphonuclear leucocytes (PMNLs), they are phagocytic and guard blood, skin and mucus membranes against infection. They respond quickly wherever tissue injury has occurred. Their azurophilic granules contain myeloperoxidases that are able to create cytotoxic substances capable of killing bacteria and other engulfed pathogens.
- Dendritic cells (DC) have long membrane extensions that resemble the dendrites of nerve cells, hence their name. These cells are phagocytic and are involved in initiating the acquired immunity.

Answer (B)

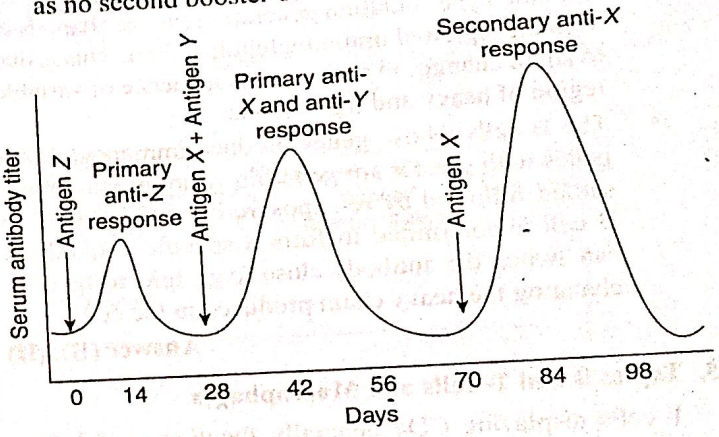
42. Topic: Antigen-Antibody Reaction

The initial encounter of a naive immunocompetent lymphocyte with an antigen induces primary response; a later contact of the host with the same antigen induce a more rapid and heightened secondary response. The amplified population of memory cells accounts for the rapidity and intensity that distinguishes a secondary response from the primary response.

The primary response has a lag of approximately 5–7 days before antibody levels start to rise. This lag is the time required for activation of naive B cells by antigen and T_H cells and for the subsequent proliferation and differentiation of the activated B cells into plasma cells. Antibody levels peak in the primary response at about day 14 and then begin to drop off as the plasma cells begin to die. In the secondary response, the lag is much shorter (only 1–2 days), antibody levels are much higher, and they are sustained for much longer. The secondary response reflects the activity of the clonally expanded population of memory B cells. These memory cells respond to the antigen more rapidly than naive B cells; in addition, because there are many more memory cells than there were naive B cells for the primary response, more plasma cells are generated in the secondary response, and antibody levels are consequently 100- to 1000-fold higher.

In the graph, on injection of Z antigen in the body for the first time, primary response of Z-specific Ig G will be at peak at day 14. By day 28, when the response against antigen Z falls out, then, on injection of antigen X and Y for the first time, primary response of X and Y-specific Ig G will be at peak at day 42 (14 days after injection of antigen X and Y). By day 70, when the primary response against X and Y would fall out, a second booster of antigen X will generate elicit secondary response against antigen X within 1–2 days and it will remain for a longer time period.

By day 84, secondary response against antigen X will be at peak, but Y-specific antibodies will be lacking as no second booster of Y antigen were given.



Answer (D)

43. Topic: Antibody Structure and Function

When the fusion of V, D and J segments due to addition or deletion of nucleotides is not precise, functional diversion of antibodies arises. It can either arise from deletion of nucleotide in the joint region or by addition of P/N nucleotide during VDJ recombination.

- P nucleotide are added when the DNA in the intervening region between gene segments is nicked as a result of which one strand on each gene segment has a recessed end while the other has an overhang. DNA repair enzymes fill the gaps with new nucleotides and gene segments are ligated.
- The N nucleotides are found in VD or DJ joints that do not appear in germline sequence.

Sometimes deletion also takes place when exonuclease trims the coding sequence eliminating a few original nucleotides. Thus, junctional diversity is one of the methods by which variability in V-J and V-D-J joining leads to antibody diversity.

Answer (B)

44. Topic: Complement

- Affinity maturation is the process by which B cells increase their affinity for a particular antigen. This affinity of B-cell specificity occurs through a repeated process of somatic hypermutation of B-cell receptors and subsequent clonal selection.
- Immunoglobulin class switching, also known as isotype switching, is a biological mechanism that changes a B cell's production of immunoglobulin (antibodies) from one type to another. During this process, the constant-region portion of the antibody heavy chain is changed, but the variable region of the heavy chain stays the same (the terms "variable" and "constant" refer to changes or lack thereof between antibodies that target different epitopes). Since the variable region does not change, class switching does not affect antigen specificity.
- Somatic hypermutation generates random changes in variable region of immunoglobulin genes. This is due to small changes in amino acids sequence of variable region of heavy and light chains.
- The B cells whose genes produce immunoglobulin genes with greater antigen affinity are preferentially settled followed by re-exposure by antigen. Once a B cell is committed to form a specific antibody, it can switch the antibody close (e.g., IgM to IgG) by changing the heavy chain produced in the cell.

Answer (B), (D)

45. Topic: B and T-Cells and Macrophages

T cells displaying CD4 generally function as T helper (T_H) cells, whereas those displaying CD8 generally function as T cytotoxic (T_C) cells.

- On interaction with an antigen–MHC class II molecule complex, T_H cell gets activated and turns into an effector cell that secretes cytokines. These cytokines play an important role in activating B cells, T_C cells, macrophages, and various other cells that participate in the immune response.
- Under the influence of T_H-derived cytokines, a T_C cell on recognition of an antigen–MHC class I molecule complex proliferates and differentiates into an effector cell called a cytotoxic T lymphocyte that exhibits cell-killing or cytotoxic activity.

Cytotoxic T cells (displaying CD8+) recognize antigens that are associated with MHC class I molecules. Since all cells express MHC class I molecules on their surface, cytotoxic T cells fight an infection irrespective of the cell affected.

Helper cells (displaying CD4+) recognize antigens that are associated with MHC class II molecules that are found mainly on dendritic cells and macrophages. Upon activation TH cells stimulate B cells to produce antibodies that bind to exogenous antigen in the body.

Answer (C)

46. Topic: Antigen Processing and Presentation

In antigenic variation, genetic recombination between two genes modifies the expressed surface antigen. It is used by human pathogens to evade host immune responses. In antigenic variation, surface antigens are changed regularly to avoid immune recognition.

Answer (C)

47. Topic: Antigen-Antibody Reaction

Enzyme-linked immunosorbent assay (ELISA) is a technique that permits detection of very small amount of protein (both as antigen and antibody) with the help of enzymes. It is an economical, rapid and highly sensitive method for screening a large number of samples simultaneously.

The following steps are required to be carried out in this test:

Step 1: Antigen bound to a well of a plastic plate reacts with the antibody being detected. The anti-antibody has an enzyme tag attached to it.

Step 2: After the antibody being measured has reacted with the antigen, the anti-antibody-enzyme complex is added.

Step 3: A substrate, specific to the enzyme, which can be converted to a colored product is applied. The amount of colored product is proportional to the concentration of the antibody.

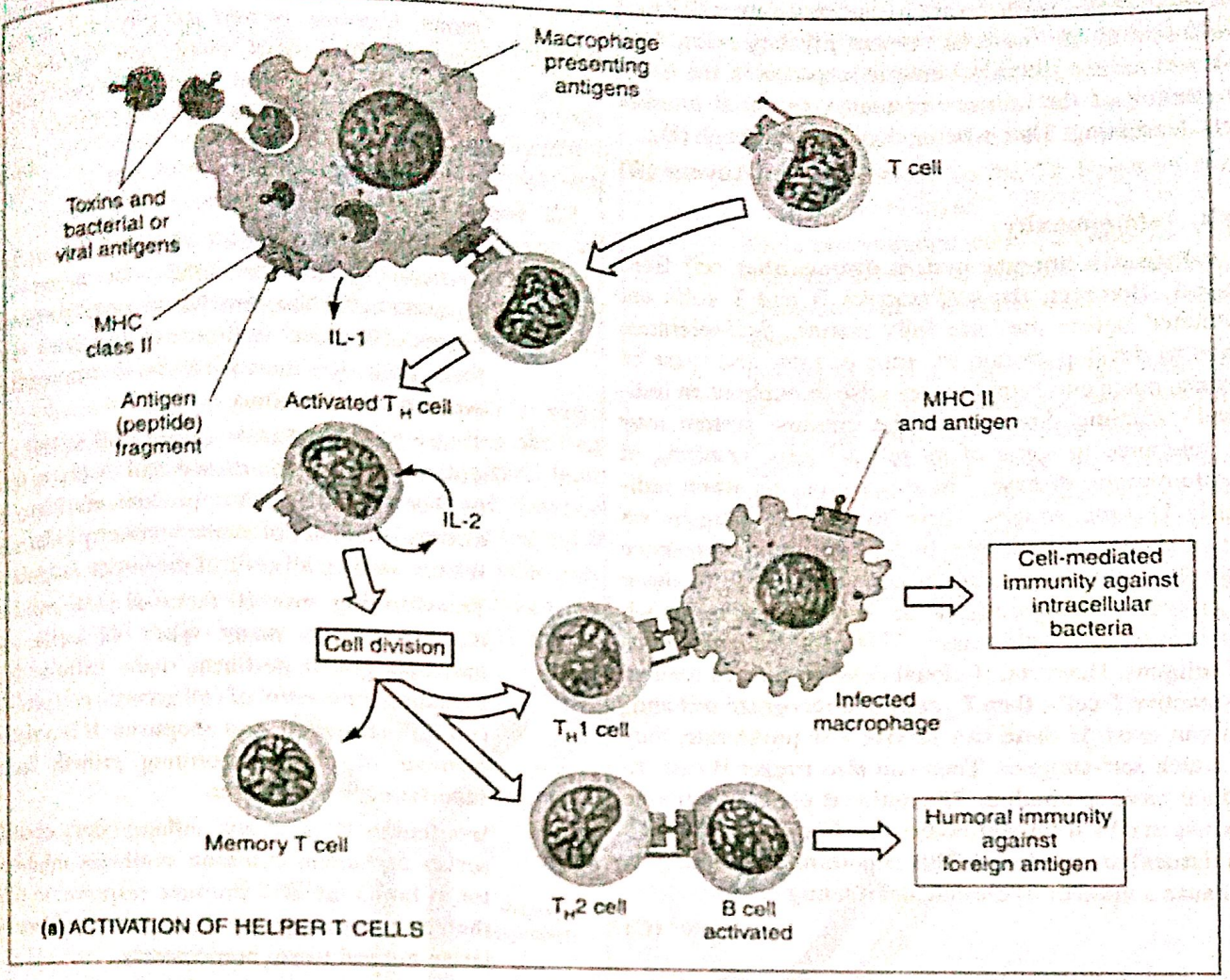
Answer (A)

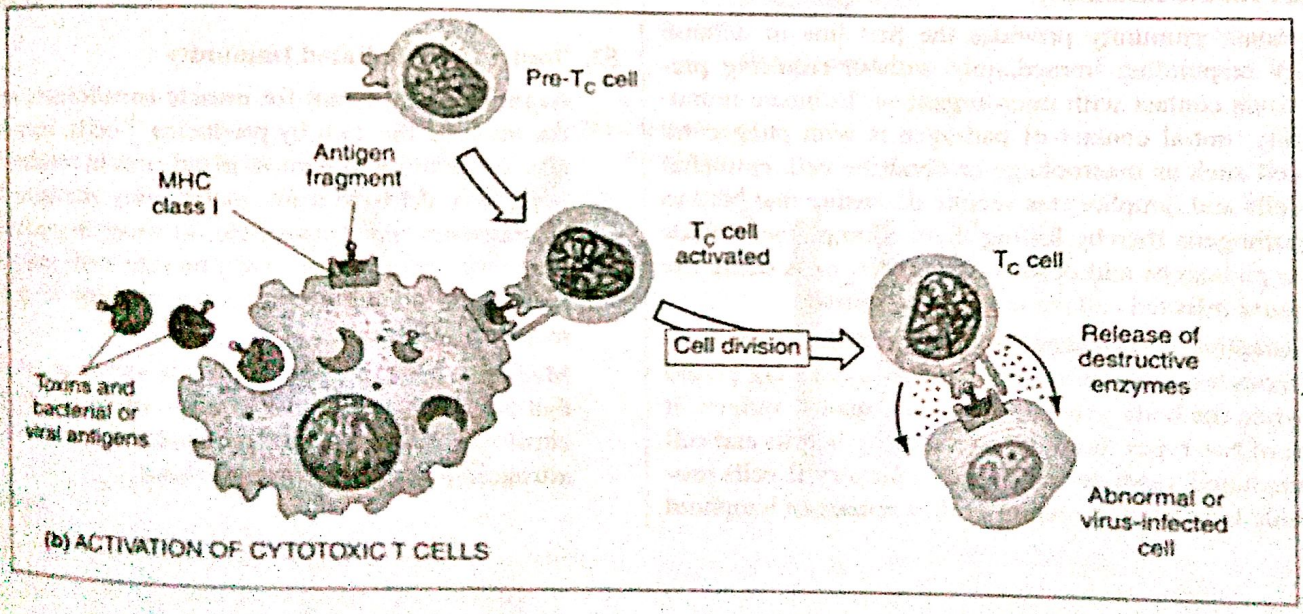
48. Topic: B and T Cells of Macrophages

When macrophages and dendritic cells phagocytize pathogens, they ingest and degrade the pathogen. Pieces of the pathogen (peptides) are then transported to the surface of these cells. Some of the pieces of the pathogen's antigen molecules are inserted into the cell membranes of antigen presenting cells. The peptide is bound to the surface of the cell by MHC II proteins. When a macrophage presents the antigen to T cells, the antigen and CD4 (T-cell receptor) bind. Once activated, T_H cells can stimulate other T and B cells, as well as phagocytes. The activated T cells then differentiate into either T_H1 cells or T_H2 cells. T_H1 cells activate infected macrophages to destroy internal bacterial infections. T_H2 cells activate B cells (humoral immune responses) by binding to MHC class II peptide presented by the B cells.

In cell-mediated immunity antigens are present on the surface of the cells along with major histocompatibility (MHC) protein that are of two types:

- MHC – Present in all nucleated cells.
- MHC II – Present in antigen presenting cells (APC) along with MHC I.





Answer (C)

49. Topic: Immune Tolerance

An immunocompetent body is able to recognize antigens and act against them. The initial encounter of an immunocompetent person with a pathogenic bacterium will cause an increase in the bacterial number till they reach their peak and induce primary immune response in the body. On starting of the immune response, bacterial number starts decreasing. This is being depicted by Graph (B).

Answer (B)

50. Topic: Autoimmunity

An individual's immune system distinguishes self from non-self. However, the self-reactive B and T cells are eliminated before they are fully mature. Self-tolerance begins to develop around the time of birth and must be ongoing, since new lymphocytes arise throughout an individual's lifetime. Sometimes, the immune system may lose tolerance to some of its self-antigens, resulting in an autoimmune disease. The disease occurs when individuals become hypersensitive to specific antigens on cells or tissues of their own bodies, despite the presence of mechanisms that generally create tolerance to these self-antigens. Autoimmunization is a process by which hypersensitivity to self occurs. CD4 cells recognize non-self antigens. However, if clonal deletion fails to remove self-reactive T cells, then T_H cells that recognize self antigens can exist. If these can survive and proliferate, they can attack self-antigens. They can also trigger B cells to produce more antibodies. The antigens elicit an immune response in which autoantibodies (antibodies against one's own tissues) are produced. This condition, autoimmunity, can cause a number of chronic debilitating diseases.

Answer (C)

51. Topic: Innate Immunity

- Innate immunity provides the first line of defense by responding immediately without requiring previous contact with microorganism. In innate immunity, initial contact of pathogen is with phagocytic cell such as macrophage or dendritic cell, epithelial cells and lymphocytes secrete defensins that bind to pathogens thereby killing them. Complement binds to pathogens and destroys them. NK cells cause the more infected cells to undergo apoptosis.
- Adaptive or acquired immunity, mediated by lymphocytes, is not immediate and requires a lag period when the body gears up for attack against antigen. It is of two types: humoral mediated by B cells and cell mediated, mediated by T cells. Memory B cells provide long term immunity or they remain in lymphoid tissues and respond rapidly if antigen reappears in the body in future.
- Memory B cells are one the principal components of the innate immunity. The major components of the innate immune system are physical epithelial barriers, dendritic cells, phagocytic leukocytes, small pathogens or a natural killer (NK) cells and circulating plasma proteins called complement.

Answer (B)

52. Topic: Innate Immunity

- Cytokines are proteins secreted by cells of the immune system that affect the behavior of the immune cells. They mediate several pathologic processes. They are endogenous pyrogens that cause fever induction increasing the thermoregulatory set point in hypothalamus.
- Tumor Necrosis Factor alpha (TNF alpha) is responsible for acute inflammation and mediates fever. TNF has been found to induce production of Interleukin-1 a central mediator of innate immunity inflammation it is expressed by all cells of the innate immune system.
- Transforming growth factor-β is a cytokine that is secreted by many types of cells including macrophages. It performs many cellular functions including the control of cell growth, cell proliferation, cell differentiation, and apoptosis. It is a polypeptide member of the transforming growth factor beta superfamily of cytokines.
- Interleukin-10 is an anti-inflammatory cytokine that serves as human cytokine synthesis inhibitory factor. It limits the host immune response to pathogens thereby preventing any damage to the host by regulating normal tissue homeostasis.

Answer (A)

53. Topic: Cell Mediated Immunity

Arginine is important for muscle metabolism, enhances the immune function by producing T-cells, increases the size and activity of thymus gland, aids in insulin production, liver detoxification, maintaining nitrogen balance by transport and storage, etc. In order to perform their functions, macrophages must be activated; macrophages require the uptake of exogenous arginine to meet their metabolic demands.

Macrophages express nitric oxide synthase, an enzyme that metabolizes arginine to nitric oxide and by product citrulline. NO is further metabolized to form reactive nitrogen species which are cytotoxic.

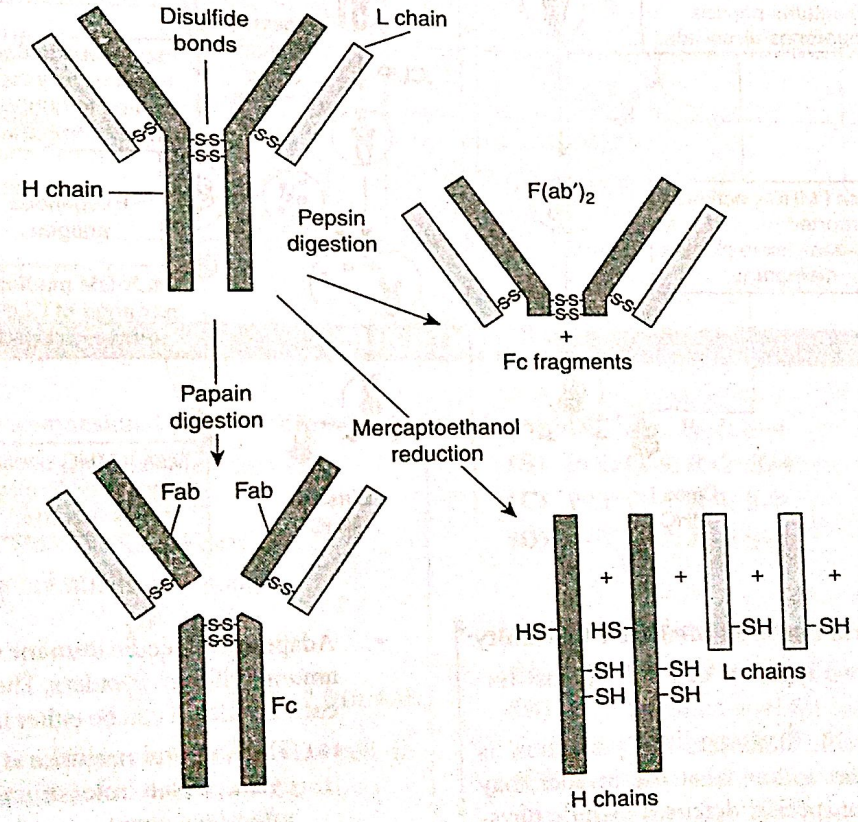
Answer (A)

54. Topic: Antibody Structure and Function

Immunoglobulin G (IgG) are found in the low molecular-weight fraction on separation to from the γ-globulin fraction of serum.

- Digestion of IgG with the enzyme papain produced three fragments:
  - (i) Two of three fragments were identical (each with a MW of 45,000) and had antigen-binding activity and were called Fab fragments ("fragment, antigen binding").
  - (ii) Third fragment (MW of 50,000) was quite different and had no antigen binding activity at all. It was found to crystallize during cold storage, thus it was called the Fc fragment ("fragment, crystallizable").
- Digestion of IgG with pepsin, generated a single 100,000 MW fragment composed of two Fab-like fragments designated the $F(ab')_2$ fragment, which binds antigen. The Fc fragment was not recovered from pepsin digestion as it was digested into multiple fragments.
- Each IgG molecule contains two 50,000 MW polypeptide chains, designated as heavy (H) chains, and two 25,000 MW chains, designated as light (L) chains.
- To determine the relation between the enzyme digestion products—Fab, $F(ab')_2$ and Fc and heavy-chain and light-chain reduction products, antisera from goat that had been previously immunized with either the Fab fragments or the Fc fragments of rabbit IgG was used. It was found that:
  - (i) The antibody to the Fab fragment could react with both the H and the L chains.
  - (ii) The antibody to the Fc fragment reacted only with the H chain.

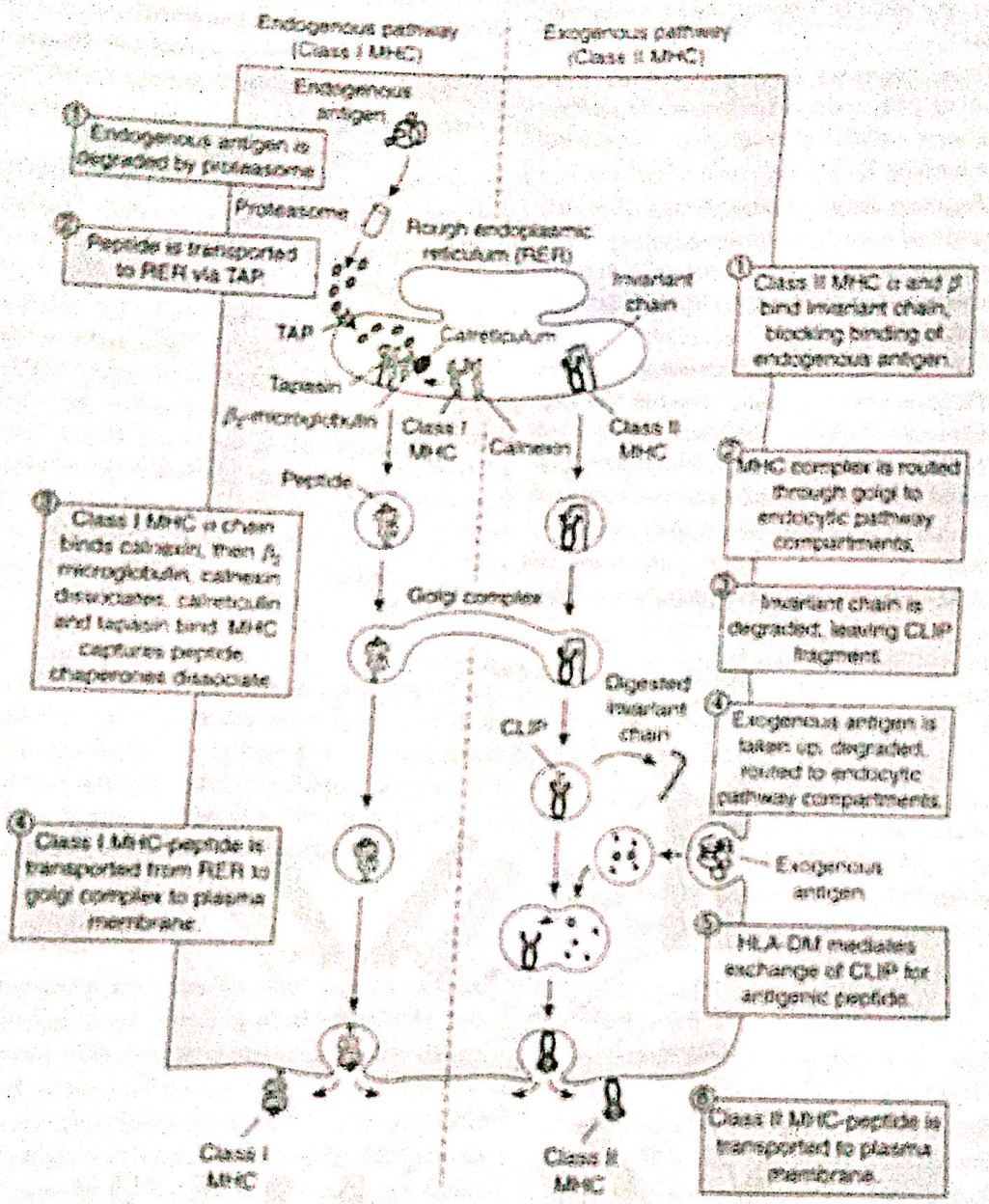
  These observations led to the conclusion that the Fab fragment consists of portions of a heavy and a light chain and that F_c contains only heavy-chain components.
- According to this model, the IgG molecule consists of two identical H chains and two identical L chains, which are linked by disulfide bridges. The enzyme papain cleaves just above the interchain disulfide bonds linking the heavy chains, whereas the enzyme pepsin cleaves just below these bonds, so that the two proteolytic enzymes generate different digestion products. Mercaptoethanol reduction and alkylation allow separation of the individual heavy and light chains.



Answer (C)

55. Topic: Major Histocompatibility Complex (MHC)

Endogenous antigens (those generated within the cell) are processed in the cytosolic pathway and presented on the membrane with class I MHC molecules, whereas exogenous antigens (those taken up by endocytosis) are processed in the endocytic pathway and presented on the membrane with class II MHC molecules.



Answer (C)

56. Topic: Innate, Humoral and Cell Mediated Immunity

In animals, there are two types of defenses against foreign invaders

- Innate or non-specific defenses, the protection is always the same, no matter what the invader may be. Examples of nonspecific defenses include physical barriers (such as skin and mucous membranes), protein defenses (interferons), cellular defenses (natural killer cells and macrophages), inflammation, and fever.
- Adaptive or specific immune responses can distinguish among different invaders. The response is different for each invader. It can be either humoral or cell-mediated.
  (a) The humoral response is mediated by B lymphocytes, which release antibodies specific to the infectious agent.
  (b) The cell-mediated response involves the binding of cytotoxic T lymphocytes to foreign or infected cells, followed by the lysis of these cells.

Answer (D)